29033327.c0

Formulation of the Multiple Premise Entailment (MPE) task

Natural Language Inference from Multiple Premises

The paper defines a new textual entailment task,
Multiple Premise Entailment (MPE), in which each
example consists of an unordered set of four
independently written image captions (premises)
and a single hypothesis sentence. The hypothesis
must be classified as entailment, neutral, or
contradiction with respect to the combined
information from all premises. This formulation
extends standard single‑premise natural language
inference by requiring aggregation across multiple
sentences, handling cross‑sentence coreference and
event resolution, and by deliberately limiting
lexical overlap between premises and hypothesis.

Prerequisite Edge Description

Description 1:
Foundational NLI resources such as SNLI, SICK, and the
RTE challenge define the
entailment/neutral/contradiction label schema, provide
benchmark data for single‑premise inference, and
establish crowdsourced annotation procedures.

Description 2:
Procedures for obtaining reliable entailment labels
via multiple annotators on platforms such as
Crowdflower and Amazon Mechanical Turk, including
majority‑vote aggregation and quality‑control
measures.

Explanation 1:
Formulating a new NLI variant requires a clear
understanding of the conventional task definition,
label set, and evaluation methodology.

Explanation 2:
Designing the MPE task required a labeling scheme and
annotation pipeline that builds on established
crowdsourcing practices used in prior NLI datasets.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Procedures for obtaining reliable entailment labels
via multiple annotators on platforms such as
Crowdflower and Amazon Mechanical Turk, including
majority‑vote aggregation and quality‑control
measures.

Explanation:
Designing the MPE task required a labeling scheme and
annotation pipeline that builds on established
crowdsourcing practices used in prior NLI datasets.

strength: strong; depth: 1

14604520.c0

Stanford Natural Language Inference (SNLI) Corpus

A large annotated corpus for learning natural
language inference

The paper introduces SNLI, a freely available
collection of 570,152 human‑written sentence pairs
labeled for entailment, contradiction, and neutral
(semantic independence). The pairs are generated
by crowdsourcing hypotheses for image captions
drawn from the Flickr30k corpus, with a balanced
three‑class distribution and a validation phase
involving multiple annotators to ensure high
quality. SNLI includes predefined train/dev/test
splits, constituency parses produced by the
Stanford PCFG parser, and is released under a
Creative Commons Attribution‑ShareAlike license,
establishing a large‑scale benchmark for
evaluating semantic representation learning and
natural language inference models.

Prerequisite Edge Description

Description:
Existing natural language inference resources such as
the Recognizing Textual Entailment (RTE) challenges
and the SICK dataset defined the
entailment/contradiction/neutral labeling scheme and
highlighted the need for larger, higher‑quality data.

Explanation:
Understanding the NLI task and its previous datasets
is essential to motivate and design a substantially
larger corpus.

strength: strong; depth: 2

16404002.c0

SICK (Sentences Involving Compositional Knowledge)
dataset

SemEval-2014 Task 1: Evaluation of Compositional
Distributional Semantic Models on Full Sentences
through Semantic Relatedness and Textual
Entailment

The authors constructed a corpus of roughly 10,000
English sentence pairs, each annotated for (i)
semantic relatedness on a 5‑point continuous scale
and (ii) textual entailment (ENTAILMENT,
CONTRADICTION, NEUTRAL). The pairs were generated
through a controlled three‑step pipeline
(normalization, expansion, pairing) to exhibit a
wide range of lexical, syntactic and semantic
compositional phenomena (e.g., active/passive
alternations, negation, operator scope) while
deliberately avoiding reliance on encyclopedic
world knowledge. The resource is released both as
a dataset for research and as a benchmark for
evaluating compositional distributional semantic
models (CDSMs) on full‑sentence semantics.

14604520.c1

Crowdsourced High-Quality NLI Data Collection
Protocol

A large annotated corpus for learning natural
language inference

The authors devise a reproducible crowdsourcing
workflow on Amazon Mechanical Turk that presents
workers with image captions from the Flickr30k
corpus as premises and asks them to write three
hypotheses—one entailing, one neutral, and one
contradictory—thereby guaranteeing a balanced
three‑class distribution. The protocol includes
detailed written instructions, an evolving FAQ,
qualification filters to exclude low‑quality
workers, and a validation stage where each pair is
re‑annotated by four additional annotators,
yielding a gold label with 98 % consensus on a
held‑out subset. By grounding each pair in a
specific visual scenario, the design mitigates
event and entity coreference ambiguities and
enforces high annotation reliability.

Prerequisite Edge Description

Description:
Theoretical work showing that ambiguous coreference
can lead to contradictory labeling decisions in
entailment tasks, and that grounding sentences in a
shared visual context can mitigate this problem.

Explanation:
The protocol’s grounding strategy and instruction
design are directly motivated by prior analyses of
coreference‑induced label ambiguity.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Theoretical work showing that ambiguous coreference
can lead to contradictory labeling decisions in
entailment tasks, and that grounding sentences in a
shared visual context can mitigate this problem.

Explanation:
The protocol’s grounding strategy and instruction
design are directly motivated by prior analyses of
coreference‑induced label ambiguity.

strength: strong; depth: 2

5555594.c3

Contradiction detection system with event‑coreference filtering

Finding Contradictions in Text

The paper implements an end‑to‑end
contradiction‑detection pipeline built on the
Stanford Recognizing Textual Entailment (RTE)
system. It adds an event‑coreference filtering
stage that discards sentence pairs that do not
describe the same event, using a root‑alignment
heuristic together with an unweighted
topicality‑based scoring model. The remaining
pairs are processed by a rich feature extractor
(polarity, numeric, antonymy, structural,
factivity, modality, relational) and classified
with a logistic‑regression model, producing a tool
for fine‑grained contradiction detection.

5555594.c5

Topicality‑based event‑coreference scoring method

Finding Contradictions in Text

The paper introduces an unweighted topicality
model that assigns higher weights to noun phrases
appearing earlier in a sentence, using their
linear position derived from typed dependency
parses. A normalized topicality score is computed
for each aligned noun‑phrase pair between a text
and a hypothesis, and combined with a
root‑alignment heuristic to decide whether the two
sentences describe the same event. This filtering
step improves precision of contradiction detection
beyond using root‑alignment alone.

207847663.c2-1

Hard-selection baseline models for rationalized
predictions

ERASER : A Benchmark to Evaluate Rationalized NLP
Models

The paper implements three hard-selection
architectures that produce discrete rationales and
use only the extracted snippets for downstream
prediction. The first adapts the Lei et al. (2016)
rationalizer, training a binary mask encoder and a
classifier decoder jointly with REINFORCE, using
either BERT or GloVe embeddings as token encoders.
The second is a pipeline that first extracts
sentence-level rationales with a GRU-based encoder
and then classifies using a separate model. The
third is a BERT-to-BERT pipeline where a BERT
encoder selects rationales and a second BERT model
consumes them for prediction. All models are
trained on the ERASER benchmark datasets and
released as open-source code.

Prerequisite Edge Description

Description:
A two-module neural model that learns a binary mask
over input tokens (rationale extractor) and a
classifier that consumes only the selected tokens,
trained jointly via REINFORCE.

Explanation:
The hard-selection baselines directly re-implement
this architecture, so understanding its design and
training objective is essential.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A deep bidirectional transformer pretrained on large
corpora, providing contextual token embeddings.

Explanation:
Two of the hard-selection baselines (Lei et al. with
BERT encoder and the BERT-to-BERT pipeline) rely on
BERT to encode inputs before rationale extraction.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Pretrained 300-dimensional word vectors learned from
global word co-occurrence statistics.

Explanation:
For datasets where BERT could not be applied (e.g.,
long documents), the hard-selection models use GloVe
embeddings as the token encoder.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Pretrained 300-dimensional word vectors learned from
global word co-occurrence statistics.

Explanation:
For datasets where BERT could not be applied (e.g.,
long documents), the hard-selection models use GloVe
embeddings as the token encoder.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A gated recurrent unit that processes token sequences
to produce sentence-level representations.

Explanation:
The pipeline baseline that extracts sentence-level
rationales uses a GRU to encode each sentence before
classification.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A first-order stochastic optimization method that
adapts learning rates for each parameter using
estimates of first and second moments of gradients.

Explanation:
All hard-selection models are trained with Adam, as
described in the hyperparameter details.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
An imperative-style, high-performance library for
building and training neural networks.

Explanation:
The implementations of all hard-selection baselines
are built using PyTorch.

strength: strong; depth: 1

7205805.c1

Modular encoder‑generator framework for rationale‑based prediction

Rationalizing Neural Predictions

The paper proposes a high‑level conceptual
framework that separates a rationale generator,
which extracts a short, coherent subset of the
input text, from an encoder that makes task
predictions using only the selected rationale. The
generator and encoder are instantiated with
bidirectional RCNNs (the generator with
independent or recurrent selection mechanisms, the
encoder as a standard RCNN classifier). The two
modules are trained jointly with a combined
prediction loss and sparsity/continuity
regularizer, using a REINFORCE‑style doubly
stochastic gradient estimator to handle the
discrete latent rationale selections. This enables
accurate predictions together with interpretable
justifications without any rationale annotations.

Prerequisite Edge Description

Description:
A neural unit that combines recurrent and
convolutional operations to capture non‑consecutive
n‑gram features, used both in the encoder and as the
base of the generator’s bidirectional network.

Explanation:
Both encoder and generator are instantiated with RCNN
layers, making this architecture a prerequisite
component.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior models that use soft or stochastic attention to
weight input tokens, providing context for why a
separate rationale generator is needed.

Explanation:
The paper positions its generator as an alternative to
attention‑based explanations, requiring awareness of
existing attention literature.

strength: strong; depth: 2

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Prior models that use soft or stochastic attention to
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Explanation:
The paper positions its generator as an alternative to
attention‑based explanations, requiring awareness of
existing attention literature.

strength: strong; depth: 2

6468765.c0

Gated non-consecutive convolutional encoder (RCNN) for question representation

Semi-supervised Question Retrieval with Gated
Convolutions

The paper introduces a recurrent convolutional
neural network (RCNN) that extends the
non‑consecutive convolutional model of Lei et al.
(2015) with an adaptive gating mechanism. At each
time step a gate λ_t, computed from the current
token embedding and the previous hidden state,
modulates how much of the accumulated n‑gram
features are retained, allowing the model to
emphasize salient words and suppress noisy tokens
in long question bodies. The sequence of gated
hidden states is pooled (mean or last state) to
produce a fixed‑dimensional question vector, and
cosine similarity between vectors is used to rank
candidate questions for retrieval.

11212020.c1

Soft attention mechanism for differentiable
alignment

Neural Machine Translation by Jointly Learning to
Align and Translate

The paper introduces a soft attention mechanism
that, for each target word, computes a context
vector as a weighted sum of encoder annotations.
The attention weights \(\alpha_{ij}\) are obtained
from a feed‑forward alignment model that scores
the compatibility between the decoder hidden state
and each source annotation, and are normalised
with a softmax. Because the whole computation is
differentiable, gradients can flow through the
alignment scores, allowing the attention
parameters, encoder, and decoder to be trained
jointly, replacing the static fixed‑length context
vector of basic encoder‑decoder models.

Prerequisite Edge Description

Description:
A sequence‑to‑sequence architecture where an encoder
RNN maps a source sentence to a representation and a
decoder RNN generates the target sentence conditioned
on that representation.

Explanation:
The attention mechanism augments this framework by
replacing the single fixed‑length representation with
a dynamic context computed from all encoder
annotations.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A sequence‑to‑sequence architecture where an encoder
RNN maps a source sentence to a representation and a
decoder RNN generates the target sentence conditioned
on that representation.

Explanation:
The attention mechanism augments this framework by
replacing the single fixed‑length representation with
a dynamic context computed from all encoder
annotations.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Gated recurrent units (GRU) or Long Short‑Term Memory
(LSTM) cells mitigate the vanishing‑gradient problem
and provide expressive hidden representations for both
encoder and decoder RNNs.

Explanation:
These hidden representations are the inputs to the
attention alignment model and the decoder state
updates.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior use of a differentiable soft attention mechanism
for handwriting synthesis, where a mixture of
Gaussians is used to weight encoder outputs.

Explanation:
Demonstrates that differentiable attention mechanisms
are feasible and informs the design of the current
attention model.

strength: strong; depth: 2

11212020.c3

Joint end‑to‑end training of attention and
translation

Neural Machine Translation by Jointly Learning to
Align and Translate

The paper introduces a training procedure that
jointly optimises the parameters of the encoder,
decoder and the soft attention (alignment) model
by maximising the conditional likelihood of the
target sentence given the source. The attention
weights \(\alpha_{ij}\) are computed by a
feed‑forward alignment network and are
incorporated as differentiable nodes in the
computational graph, allowing gradients to flow
from the translation loss through the alignment
scores back to the encoder and decoder. This
unified objective replaces the traditional
two‑stage approach where alignment is treated as a
latent variable and learned separately.

17195923.c0

Recurrent Attention Model (RAM) architecture

Recurrent Models of Visual Attention

The paper introduces a novel recurrent neural
network architecture for visual processing that
sequentially selects fixation locations, extracts
a multi‑resolution retinal glimpse at each
location, and integrates these glimpses over time
with a core RNN. The hidden state of the RNN feeds
a stochastic location network that chooses the
next fixation and an action network that produces
task‑specific outputs such as classification or
control. Because computation is limited to the
glimpses, the amount of processing is independent
of the full image size while still providing
translation invariance.

17195923.c2

End-to-end training of RAM with REINFORCE, learned baseline, and hybrid supervised loss

Recurrent Models of Visual Attention

The paper introduces a training procedure for the
Recurrent Attention Model (RAM) that combines
standard back‑propagation for differentiable
components (glimpse, core, and action networks)
with a REINFORCE policy‑gradient update for the
stochastic location network. To reduce the high
variance of the REINFORCE estimator, a learned
baseline (value function) is trained jointly with
the model. For classification tasks the final
action output is also trained with a supervised
cross‑entropy loss, yielding a hybrid objective
that leverages both reinforcement and supervised
signals.

52967399.c7

Open-source release of BERT code and pretrained checkpoints

BERT: Pre-training of Deep Bidirectional
Transformers for Language Understanding

The authors publicly release a TensorFlow
implementation of BERT together with the
pretrained BERT‑BASE and BERT‑LARGE model
checkpoints via a GitHub repository. The release
includes all source files needed to reproduce the
model architecture, pre‑training procedures, and
fine‑tuning scripts. By providing the code and
weights, the work enables reproducibility, rapid
prototyping, and further research by the community
without requiring re‑training from scratch.

1957433.c0

GloVe: Global Log‑Bilinear Regression Model for Word Embeddings

GloVe: Global Vectors for Word Representation

The paper introduces GloVe, a novel word‑embedding
model that directly factorizes a global word‑word
co‑occurrence matrix using a weighted log‑bilinear
regression objective. By modeling the logarithm of
co‑occurrence counts with a weighted least‑squares
loss, the method combines the statistical
efficiency of global matrix‑factorization
approaches with the ability of local
context‑window models to capture linear
substructures such as analogies. The resulting
vectors encode semantic and syntactic regularities
and achieve state‑of‑the‑art performance on
analogy, similarity, and named‑entity‑recognition
benchmarks.

Prerequisite Edge Description

Description:
Understanding how to build a word‑word co‑occurrence
matrix, the distributional properties of co‑occurrence
counts, and prior techniques for normalising and
weighting these counts (e.g., LSA, HAL, COALS, PPMI).

Explanation:
GloVe’s objective is defined directly on the log of
co‑occurrence counts, so knowledge of how such
matrices are constructed and interpreted is essential
for formulating the model.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Evaluating word embeddings by solving analogies of the
form a:b :: c:d using vector arithmetic (b − a + c)
and cosine similarity.

Explanation:
The motivation for GloVe’s design is to produce
embeddings that excel on this analogy task, so
understanding the evaluation method is required to
formulate the model’s objectives.

strength: strong; depth: 2

12730203.c6

Explicit high‑dimensional sparse word vectors from Wikipedia using PPMI

Linguistic Regularities in Sparse and Explicit
Word Representations

The authors build a traditional distributional
representation by counting word–context
co‑occurrences within a symmetric window of two
tokens on the English Wikipedia corpus. They
weight the raw counts with positive pointwise
mutual information (PPMI) to obtain a |V|×|C|
sparse matrix where each dimension corresponds to
a positional context (e.g., "-2", "+1"). This
explicit vector space serves as a reproducible
baseline for relational similarity experiments and
demonstrates that large‑scale sparse vectors can
be constructed from publicly available data.

7478738.c1

Vector offset algorithm for word analogy solving

Linguistic Regularities in Continuous Space Word
Representations

The paper introduces a simple algorithm that
solves word analogy questions of the form "a is to
b as c is to ?" by computing the vector y = xb –
xa + xc, normalising the resulting vector to unit
length, and selecting the vocabulary word whose
embedding has the highest cosine similarity to y.
The method directly exploits the empirically
observed constant offset property of word
embeddings learned by a recurrent neural network
language model. It provides a fast,
language‑independent reasoning mechanism that can
be applied to both syntactic and semantic analogy
sets.

1957433.c8

Open‑Source Implementation and Pre‑trained Word
Vectors Release

GloVe: Global Vectors for Word Representation

The authors publicly release the software for
training the GloVe model together with pretrained
word‑vector embeddings learned on several large
corpora (Wikipedia 2010/2014, Gigaword 5, Common
Crawl). The code, scripts, and vector files are
hosted at http://nlp.stanford.edu/projects/glove/,
enabling reproducibility and immediate downstream
use by the research community.

5590763.c1

Gated Recurrent Unit (reset and update gates)

Learning Phrase Representations using RNN
Encoder–Decoder for Statistical Machine
Translation

The paper introduces a novel recurrent hidden unit
that augments the standard RNN state with two
multiplicative gates: a reset gate r that controls
how much of the previous hidden state is ignored
when computing the candidate activation, and an
update gate z that interpolates between the
previous hidden state and the candidate
activation, thereby adaptively remembering or
forgetting information. The unit’s equations
(5)–(8) are computationally simpler than the LSTM
while providing comparable adaptive memory
capabilities. This gated unit is employed in both
the encoder and decoder of the proposed RNN
Encoder–Decoder architecture.

Prerequisite Edge Description

Description:
Prior work on leaky‑integrator units and adaptive
gating in recurrent networks, which showed that
multiplicative gates can improve memory capacity and
training stability.

Explanation:
The paper positions its unit as an adaptive variant of
a leaky‑integration unit, citing earlier work to
justify the design choice.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
An adaptive learning‑rate method for stochastic
gradient descent that reduces the need for manual
learning‑rate tuning.

Explanation:
Training the gated unit efficiently required a robust
optimizer; the paper uses AdaDelta for this purpose.

strength: strong; depth: 2

12485056.c1

Leaky‑integration hidden units for longer temporal dependencies

Advances in optimizing recurrent networks

The paper augments standard recurrent neural
networks by assigning a subset of hidden units
leaky‑integration dynamics: ht,i = αi ht−1,i + (1
− αi) F_i(ht−1,xt), where αi is sampled uniformly
from (0.02, 0.2). These slower‑changing units act
as low‑pass filters, providing shorter effective
gradient paths that mitigate vanishing gradients
and allow information to persist over longer
time‑scales. Experiments on polyphonic music and
Penn Treebank language modeling demonstrate lower
training and test loss compared to vanilla RNNs.

7365802.c0

ADADELTA adaptive learning‑rate algorithm

ADADELTA: An Adaptive Learning Rate Method

ADADELTA is a per‑dimension adaptive learning‑rate
method that updates parameters using only
first‑order gradient information. It maintains an
exponentially decaying average of squared
gradients (RMS[g]) for the denominator and an
exponentially decaying average of past parameter
updates (RMS[Δx]) for the numerator, yielding the
unit‑consistent update Δx = - (RMS[Δx]_{t-1} /
RMS[g]_{t}) * g_t. The algorithm removes the need
for a manually tuned global learning rate, adds
only trivial overhead to vanilla SGD, and is
robust across network architectures, data
modalities, and distributed training settings.

6628106.c0

Adam: Adaptive Moment Estimation Optimization Algorithm

Adam: A Method for Stochastic Optimization

Adam is a first‑order stochastic optimization
method that computes per‑parameter learning rates
using exponential moving averages of the gradient
(first moment) and the squared gradient (second
raw moment). The algorithm incorporates
bias‑corrected moment estimates to eliminate
initialization bias, yields updates invariant to
diagonal rescaling of gradients, and bounds the
effective step size by a global stepsize
hyper‑parameter. Designed for non‑stationary
objectives and sparse or noisy gradients, Adam
requires only minimal additional memory and
computation beyond standard stochastic gradient
descent.

202786778.c0

Imperative define‑by‑run deep learning library (PyTorch)

PyTorch: An Imperative Style, High-Performance
Deep Learning Library

The paper introduces PyTorch, a Python library
that performs immediate (eager) execution of
tensor operations while providing reverse‑mode
automatic differentiation and GPU acceleration. By
embracing Python's imperative programming model,
models, data loaders, and optimizers can be
written as ordinary Python code, enabling
straightforward debugging and rapid
experimentation. The authors demonstrate that
careful engineering allows PyTorch to achieve
performance comparable to static‑graph frameworks,
challenging the belief that dynamic execution
necessarily incurs a large speed penalty.

Prerequisite Edge Description

Description:
High‑performance kernels and libraries (CUDA, cuDNN,
cuBLAS) that implement convolution, matrix
multiplication, and other tensor operations on NVIDIA
GPUs.

Explanation:
To achieve performance comparable to static‑graph
frameworks, PyTorch must offload heavy computation to
optimized GPU kernels.

strength: strong; depth: 2

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Explanation:
To achieve performance comparable to static‑graph
frameworks, PyTorch must offload heavy computation to
optimized GPU kernels.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Earlier deep‑learning libraries that demonstrated the
feasibility of eager execution, albeit with different
trade‑offs in performance or language choice,
providing conceptual motivation for an imperative,
eager‑execution system in Python.

Explanation:
These works establish the feasibility of define‑by‑run
execution and highlight the need for a
high‑performance Python‑centric solution.

strength: strong; depth: 2

12330432.c0

cuDNN: a framework‑agnostic deep‑learning primitives library

cuDNN: Efficient Primitives for Deep Learning

The paper introduces cuDNN, a C‑language library
that implements highly optimized GPU kernels for
the core operations used in deep‑learning models
(convolution, pooling, activation functions,
softmax, and tensor transformations). The library
is deliberately framework‑agnostic: it exposes a
lightweight, state‑free, thread‑safe API built
around opaque descriptors that allow any tensor
layout to be passed in without copying data. cuDNN
achieves high performance and low auxiliary memory
usage by mapping convolutions onto NVIDIA's
high‑throughput GEMM engine with a lazy im2col
strategy and by applying
integer‑division‑to‑multiply indexing
optimizations.

12330432.c4-1

GPU‑accelerated forward and backward pooling primitives

cuDNN: Efficient Primitives for Deep Learning

cuDNN provides highly optimized GPU kernels for
both max and average pooling operations, including
forward passes that compute pooled outputs and
backward passes that propagate gradients with
respect to the input data. The primitives support
arbitrary 4‑D tensor layouts via stride
descriptors and require no auxiliary memory,
enabling seamless integration into existing
deep‑learning frameworks.

12330432.c4-2

GPU‑accelerated activation function primitives (sigmoid, ReLU, tanh)

cuDNN: Efficient Primitives for Deep Learning

cuDNN implements forward and backward kernels for
three widely used activation functions—sigmoid,
rectified linear unit (ReLU), and hyperbolic
tangent (tanh). The kernels respect arbitrary
tensor layouts, require no extra memory, and are
optimized for high throughput on NVIDIA GPUs.

12330432.c4-3

GPU‑accelerated numerically stable softmax primitive

cuDNN: Efficient Primitives for Deep Learning

cuDNN offers a softmax routine that computes the
softmax function in a numerically stable way by
subtracting the maximum input value before
exponentiation, with an optional
fast‑but‑less‑accurate mode. The implementation
supports forward and backward passes, arbitrary
tensor layouts, and operates without auxiliary
memory.

12330432.c4-4

Tensor transformation utilities (addition with broadcasting and transposition)

cuDNN: Efficient Primitives for Deep Learning

cuDNN includes auxiliary tensor operations such as
element‑wise addition with optional broadcasting
and tensor transposition. These utilities respect
arbitrary stride‑based layouts, require no extra
memory, and are designed to be called from
deep‑learning frameworks to simplify common tensor
manipulations.

2170930.c1

DyNet toolkit implementing dynamic declaration

DyNet: The Dynamic Neural Network Toolkit

DyNet is an open‑source C++ library with Python
bindings that embodies the dynamic declaration
paradigm, constructing the computation graph
implicitly during the forward pass of each
training instance. It provides a lightweight graph
representation, custom memory pools for forward
and backward values, and an optimized backend that
makes per‑example graph construction fast enough
for practical training. The toolkit is released
under the Apache 2.0 license and is designed to be
idiomatic in both C++ and Python, enabling rapid
prototyping of neural models with minimal
boilerplate.

207847663.c2-2

Soft‑selection baseline models for rationalized
predictions

ERASER : A Benchmark to Evaluate Rationalized NLP
Models

The paper implements a suite of soft‑selection
baselines that assign continuous importance scores
to each token. A BERT encoder followed by a
bidirectional LSTM produces contextual token
representations; token‑level attention,
gradient‑based saliency, and LIME are applied to
obtain soft importance scores. These scores are
evaluated for agreement with human rationales and
for faithfulness using the ERASER metrics. The
models are trained on the ERASER benchmark and the
implementation is released as open‑source code.

Prerequisite Edge Description

Description:
A deep bidirectional transformer that yields
contextual token embeddings for downstream tasks.

Explanation:
The soft‑selection models use BERT as the first
encoder layer to obtain contextualized token
representations.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A deep bidirectional transformer that yields
contextual token embeddings for downstream tasks.

Explanation:
The soft‑selection models use BERT as the first
encoder layer to obtain contextualized token
representations.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A soft alignment method that computes a weighted sum
of hidden states using learned attention scores,
originally proposed for neural machine translation.

Explanation:
One soft‑selection baseline computes token importance
via attention over the LSTM hidden states.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
An attribution method that integrates gradients of the
model output with respect to inputs along a path from
a baseline to the actual input.

Explanation:
The gradient baseline computes token‑level importance
scores using this technique.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A model‑agnostic technique that approximates the local
decision boundary of a classifier with an
interpretable surrogate model to obtain feature
importance.

Explanation:
One soft‑selection baseline uses LIME to generate
token importance scores for each instance.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A model‑agnostic technique that approximates the local
decision boundary of a classifier with an
interpretable surrogate model to obtain feature
importance.

Explanation:
One soft‑selection baseline uses LIME to generate
token importance scores for each instance.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A stochastic optimization algorithm that adapts
learning rates per parameter using estimates of first
and second moments of gradients.

Explanation:
All soft‑selection models are trained with Adam as
described in the hyperparameter section.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A stochastic optimization algorithm that adapts
learning rates per parameter using estimates of first
and second moments of gradients.

Explanation:
All soft‑selection models are trained with Adam as
described in the hyperparameter section.

strength: strong; depth: 1

Prerequisite Edge Description

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A stochastic optimization algorithm that adapts
learning rates per parameter using estimates of first
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A stochastic optimization algorithm that adapts
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described in the hyperparameter section.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A stochastic optimization algorithm that adapts
learning rates per parameter using estimates of first
and second moments of gradients.

Explanation:
All soft‑selection models are trained with Adam as
described in the hyperparameter section.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A stochastic optimization algorithm that adapts
learning rates per parameter using estimates of first
and second moments of gradients.

Explanation:
All soft‑selection models are trained with Adam as
described in the hyperparameter section.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
An imperative‑style, high‑performance library for
building and training neural networks.

Explanation:
The soft‑selection baselines are implemented using
PyTorch.

strength: strong; depth: 1

52967399.c0

Bidirectional Transformer encoder architecture
(BERT)

BERT: Pre-training of Deep Bidirectional
Transformers for Language Understanding

BERT introduces a multi‑layer Transformer encoder
in which every token attends to both its left and
right context through fully bidirectional
self‑attention. The model combines WordPiece token
embeddings, learned segment embeddings, positional
embeddings, and special tokens ([CLS], [SEP]) to
represent single sentences or sentence pairs
uniformly. This architecture serves as a universal
language representation that can be fine‑tuned
with a minimal task‑specific output layer for a
wide range of NLP tasks.

Prerequisite Edge Description

Description:
The Transformer encoder introduced by Vaswani et al.
(2017) uses multi‑head self‑attention, positional
embeddings, and feed‑forward sub‑layers to process
sequences in parallel, enabling deep contextual
representations.

Explanation:
BERT directly builds on the Transformer encoder,
replacing the unidirectional decoder with a fully
bidirectional encoder.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
WordPiece splits words into subword units based on a
learned vocabulary, allowing open‑vocabulary handling
and efficient embedding lookup.

Explanation:
BERT’s input representation sums token, segment, and
position embeddings derived from WordPiece tokens.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior work (e.g., OpenAI GPT, ELMo) showed that
left‑to‑right or right‑to‑left language models cannot
condition on both sides simultaneously, limiting
performance on tasks that require full context.

Explanation:
Motivates the need for a bidirectional pre‑training
architecture, which BERT provides.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior work (e.g., OpenAI GPT, ELMo) showed that
left‑to‑right or right‑to‑left language models cannot
condition on both sides simultaneously, limiting
performance on tasks that require full context.

Explanation:
Motivates the need for a bidirectional pre‑training
architecture, which BERT provides.

strength: strong; depth: 2

13756489.c0

Transformer: attention‑only encoder‑decoder
architecture

Attention is All you Need

The paper presents the Transformer, a novel
sequence‑to‑sequence model that replaces recurrent
and convolutional layers with stacks of multi‑head
self‑attention and position‑wise feed‑forward
networks in both encoder and decoder. Positional
information is injected via sinusoidal encodings,
and the model employs residual connections, layer
normalization, weight tying, dropout, label
smoothing, and a warm‑up inverse‑sqrt
learning‑rate schedule. This architecture achieves
state‑of‑the‑art translation quality while
dramatically reducing training time.

3603249.c3

Wordpiece sub‑word segmentation for open‑vocabulary NMT

Google's Neural Machine Translation System:
Bridging the Gap between Human and Machine
Translation

GNMT adopts a data‑driven wordpiece model that
learns a fixed vocabulary of 8 k–32 k sub‑word
units shared between source and target languages.
The model uses a greedy algorithm to select
wordpieces that maximize language‑model likelihood
on the training corpus, yielding a deterministic
segmentation for any word. This representation
balances the flexibility of character models with
the efficiency of word‑level models, enabling
efficient handling of rare and out‑of‑vocabulary
words while keeping sequence lengths manageable
for the deep LSTM encoder‑decoder.

3626819.c0

ELMo: Deep Contextualized Word Representations

Deep contextualized word representations

The paper introduces ELMo (Embeddings from
Language Models), a novel class of word
representations that are computed as functions of
the entire input sentence rather than fixed type
embeddings. ELMo vectors are derived from the
internal states of a deep bidirectional LSTM
language model that is pretrained on a large
monolingual corpus, enabling the representations
to capture both complex syntactic/semantic
characteristics and context‑dependent polysemy.
The authors show that ELMo can be concatenated to
any existing NLP architecture with minimal
changes, providing a universal source of
linguistic knowledge that improves
state‑of‑the‑art performance on a wide range of
downstream tasks. This contribution establishes a
conceptual framework for using deep contextualized
representations as a general paradigm in NLP.

3626819.c1

Pretrained Bidirectional Language Model
Architecture

Deep contextualized word representations

The authors design a two‑layer bidirectional LSTM
language model that incorporates character‑level
convolutional token embeddings, two highway
layers, and a residual connection between the LSTM
layers. The model is trained on the 1‑Billion Word
Benchmark for ten epochs, achieving a
forward‑backward perplexity of 39.7 while
balancing capacity and computational cost. This
pretrained biLM serves as the backbone for
generating deep contextualized ELMo word
representations, and the pretrained weights and
extraction code are released publicly for reuse.

5590763.c0

RNN Encoder–Decoder architecture for
sequence‑to‑sequence learning

Learning Phrase Representations using RNN
Encoder–Decoder for Statistical Machine
Translation

The paper introduces the RNN Encoder–Decoder, a
neural network composed of two recurrent neural
networks: an encoder that reads an input sequence
of arbitrary length and compresses it into a
fixed‑dimensional vector, and a decoder that
generates an output sequence conditioned on this
vector. The encoder and decoder are trained
jointly to maximize the conditional log‑likelihood
of target sequences given source sequences,
enabling both scoring of source‑target phrase
pairs and generation of target sequences. This
design extends earlier fixed‑size feed‑forward
translation models to truly variable‑length inputs
and outputs, providing a new way to represent
whole sequences in a continuous latent space.

7961699.c0

Encoder–decoder LSTM architecture for general sequence-to-sequence learning

Sequence to Sequence Learning with Neural Networks

The paper introduces a domain‑independent,
end‑to‑end framework that maps an input sequence
to a fixed‑dimensional vector with a four‑layer
Long Short‑Term Memory (LSTM) encoder and
generates the output sequence with a separate
four‑layer LSTM decoder. The model is trained with
stochastic gradient descent using back‑propagation
through time and a softmax output over a large
vocabulary, eliminating the need for hand‑crafted
alignment models and enabling direct translation
of variable‑length sentences.

1697424.c6

Soft‑window attention mechanism for conditional
handwriting synthesis

Generating Sequences With Recurrent Neural
Networks

Introduces a differentiable soft‑window layer that
convolves a mixture of Gaussian kernels with a
one‑hot encoded character sequence, producing a
context vector fed to each LSTM layer at every
timestep. The window parameters (center location,
width, and importance) are predicted by the
network itself, allowing the model to learn an
implicit alignment between characters and
pen‑stroke positions during generation. Integrated
with a deep stacked LSTM‑MDN architecture, this
mechanism enables conditional generation of
realistic online handwriting from arbitrary text
strings in diverse styles.

16747630.c1

Integrated Gradients attribution method

Axiomatic Attribution for Deep Networks

The paper introduces Integrated Gradients, a
feature attribution technique that computes the
path integral of a network's gradients along the
straight‑line segment from a baseline input to the
actual input. The integral is approximated with a
Riemann sum (e.g., 50 steps) using only standard
gradient calls, so no network modification is
required. Integrated Gradients provably satisfies
the Sensitivity and Implementation Invariance
axioms as well as the Completeness property, and
can be implemented in any deep‑learning framework
such as TensorFlow.

Prerequisite Edge Description

Description:
Using the gradient of the output with respect to the
input as a primitive attribution signal, originally
proposed for visualizing neural‑network decisions.

Explanation:
Integrated Gradients builds directly on the gradient
operator; understanding its behavior is essential for
formulating the new method.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Using the gradient of the output with respect to the
input as a primitive attribution signal, originally
proposed for visualizing neural‑network decisions.

Explanation:
Integrated Gradients builds directly on the gradient
operator; understanding its behavior is essential for
formulating the new method.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Using the gradient of the output with respect to the
input as a primitive attribution signal, originally
proposed for visualizing neural‑network decisions.

Explanation:
Integrated Gradients builds directly on the gradient
operator; understanding its behavior is essential for
formulating the new method.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Knowledge of existing attribution techniques
(DeepLift, LRP, Deconvolutional networks, Guided
back‑propagation) and how each fails to satisfy at
least one of the proposed axioms.

Explanation:
Identifying the shortcomings of these methods
motivated the need for a new approach that satisfies
both axioms.

strength: strong; depth: 2

1450294.c1

Image-specific class saliency map computation via single back-propagation

Deep Inside Convolutional Networks: Visualising
Image Classification Models and Saliency Maps

The paper introduces a fast procedure to obtain a
class‑specific saliency map for a given image by
back‑propagating the gradient of the class score
with respect to the input pixels. The absolute
value of the gradient at each pixel (or the
maximum across colour channels) quantifies the
pixel's influence on the class score, yielding a
spatial support map in a single backward pass. The
method requires only image‑level class labels and
no additional supervision, enabling rapid visual
explanation of ConvNet decisions.

1450294.c2

Weakly supervised object localisation using class saliency maps and GraphCut segmentation

Deep Inside Convolutional Networks: Visualising
Image Classification Models and Saliency Maps

The paper proposes a pipeline that localises
objects in images using only image-level class
labels. An image-specific class saliency map is
first obtained by back-propagating the gradient of
the class score with respect to the input pixels.
The saliency map is thresholded at high and low
quantiles to initialise foreground and background
colour Gaussian Mixture Models, which drive an
interactive GraphCut segmentation to produce a
full object mask; a bounding box is then extracted
from this mask. The method is evaluated on the
ILSVRC‑2013 localisation challenge, achieving
46.4% top‑5 localisation error without any
bounding‑box supervision.

1450294.c4-2

Empirical evaluation of weakly supervised object localisation using class saliency maps and GraphCut

Deep Inside Convolutional Networks: Visualising
Image Classification Models and Saliency Maps

The authors compute image‑specific class saliency
maps by a single back‑propagation pass, threshold
them, and initialise colour‑based Gaussian mixture
models for foreground and background. An
interactive GraphCut segmentation refines the
mask, from which the largest connected component
yields a bounding box. Applied to the ILSVRC‑2013
localisation challenge, the pipeline achieves 46.4
% top‑5 localisation error without any
bounding‑box supervision, demonstrating the
practicality of gradient‑based saliency for weakly
supervised detection.

8564234.c0

DeepLIFT reference‑based feature attribution
method

Not Just a Black Box: Learning Important Features
Through Propagating Activation Differences

DeepLIFT introduces a reference‑based attribution
technique that computes importance scores by
comparing each neuron's activation to a reference
activation and propagating contribution scores
backward using specially derived multipliers. It
formalizes two core properties—summation‑to‑delta
(contributions sum to the output difference from
the reference) and linear composition
(contributions distribute proportionally through
downstream connections)—which overcome
zero‑gradient issues of traditional saliency
methods. The method provides a unified algorithm
applicable to affine layers, max‑pooling, and
common nonlinearities, and can be implemented
efficiently on GPUs.

13029170.c0

LIME: Local Interpretable Model‑agnostic Explanations

"Why Should I Trust You?" Explaining the
Predictions of Any Classifier

LIME introduces a model‑agnostic explanation
technique that approximates any black‑box
classifier locally with an interpretable linear
model defined over a human‑understandable
representation (e.g., binary word presence). It
generates perturbed samples around the instance of
interest, weights them with an exponential kernel
based on cosine distance, and fits a sparse linear
model by selecting a small set of features via
Lasso or forward selection and estimating their
weights with weighted least squares. The resulting
explanation highlights the most influential
interpretable components, providing a faithful and
intelligible rationale for the classifier’s
prediction.

13029170.c2

Open‑source LIME Software and Demonstration System

"Why Should I Trust You?" Explaining the
Predictions of Any Classifier

The authors release a fully documented Python
package that implements the LIME and SP‑LIME
algorithms and provides an interactive
Jupyter‑notebook demo. The demo integrates the
explanation engine with three NLP classification
tasks (politeness detection, 20‑newsgroups
document classification, and sentiment analysis),
allowing users to input free‑form text, view
highlighted explanatory words, compare multiple
models, and iteratively perform feature cleaning
to improve classifiers. All code and documentation
are publicly available on GitHub, enabling reuse
and further research.

Prerequisite Edge Description

Description:
Publicly available corpora for politeness detection,
20‑newsgroups document classification, and sentiment
analysis that provide realistic text inputs for
evaluating explanations.

Explanation:
The demonstration showcases the explanation system on
these datasets; access to and understanding of the
data formats are required to set up the demo.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Publicly available corpora for politeness detection,
20‑newsgroups document classification, and sentiment
analysis that provide realistic text inputs for
evaluating explanations.

Explanation:
The demonstration showcases the explanation system on
these datasets; access to and understanding of the
data formats are required to set up the demo.

strength: strong; depth: 2

12383721.c0

Large-scale politeness‑annotated request corpus

A computational approach to politeness with
application to social factors

The authors harvested over 10,000 user‑to‑user
request utterances from Wikipedia talk pages and
Stack Exchange comments, each consisting of
exactly two sentences with the second being a
request. Each request was annotated for perceived
politeness by five independent Amazon Mechanical
Turk workers, with worker‑level z‑score
normalization and extensive vetting (U.S.
residence restriction, linguistic background
questionnaire, paraphrasing attention check). The
averaged politeness scores constitute the largest
publicly released corpus of requests with
fine‑grained politeness annotations, supporting
downstream linguistic analysis and classifier
training.

12383721.c7

Public release of politeness detection dataset and
software

A computational approach to politeness with
application to social factors

The authors make publicly available the large
politeness‑annotated request corpus together with
the feature‑extraction scripts and the trained
support‑vector‑machine classifier. This release
enables other researchers to replicate the
experiments, benchmark new models, and apply
politeness detection to novel online communication
data. The resources are provided as a reusable
dataset and as open‑source software tools hosted
online.

6628106.c1

Bias‑Correction for Exponential Moving Averages in
Adam

Adam: A Method for Stochastic Optimization

The authors observe that the exponential moving
averages used to estimate the first (mean) and
second (uncentered variance) moments of stochastic
gradients are biased toward zero when initialized
with zero vectors. They derive closed‑form
correction factors that divide the raw moment
estimates by (1 − β₁^t) and (1 −β₂^t),
respectively, yielding unbiased moment estimates
even for decay rates close to one or highly sparse
gradients. The corrected estimates are
incorporated into the Adam update rule and
experimentally shown to improve stability and
convergence.

6628106.c2-1

O(√T) Regret Bound for Adam

Adam: A Method for Stochastic Optimization

The paper proves that, under the online convex
optimization setting with convex loss functions,
bounded gradients, a stepsize schedule α_t ∝ 1/√t,
and exponentially decaying first‑moment
coefficients β1,t, the Adam optimizer attains a
regret bounded by O(√T). The proof leverages
bias‑corrected moment estimates and adapts
techniques from adaptive gradient methods to
establish the sublinear regret guarantee.

6628106.c2-2

Average Regret Convergence to Zero

Adam: A Method for Stochastic Optimization

As a corollary of the O(√T) regret bound, the
paper shows that the average regret of Adam,
defined as the cumulative regret divided by T,
converges to zero as T → ∞, establishing
asymptotic optimality of the algorithm for convex
stochastic optimization.

6628106.c5-1

Invariance of Adam's Effective Step Size to Diagonal Rescaling of Gradients

Adam: A Method for Stochastic Optimization

The paper proves that Adam's update rule yields an
effective step size that is unchanged when the
gradient vector is multiplied element‑wise by any
positive diagonal matrix. By examining the
bias‑corrected first‑moment estimate m̂_t and
second‑moment estimate v̂_t, the authors show that
scaling each coordinate of the gradient by a
constant c scales m̂_t by c and v̂_t by c², which
cancel in the ratio m̂_t/√v̂_t, leaving the
parameter update magnitude invariant. This
property explains Adam's robustness to differing
gradient magnitude scales across parameters.

6628106.c5-2

Upper Bound on Per‑Parameter Update Magnitude Proportional to Global Stepsize α

Adam: A Method for Stochastic Optimization

The authors prove that each parameter update
produced by Adam satisfies |Δ_t| ≤ α, where α is
the global stepsize hyper‑parameter. By
substituting the bias‑corrected first‑moment
estimate \hat{m}_t and second‑moment estimate
\hat{v}_t into the effective stepsize formula Δ_t
= α·\hat{m}_t / (√\hat{v}_t + ε) and using the
fact that the (uncentered) second moment always
dominates the squared first moment, they show that
the ratio is bounded by 1. This yields a
trust‑region‑like guarantee that the step size
cannot exceed the prescribed α, simplifying
hyper‑parameter selection.

6628106.c5-3

Signal-to-Noise Ratio Interpretation and Automatic Annealing of Step Sizes

Adam: A Method for Stochastic Optimization

The paper interprets the bias‑corrected
first‑moment estimate \(\hat{m}_t\) divided by the
square‑root of the bias‑corrected second‑moment
estimate \(\sqrt{\hat{v}_t}\) as a signal‑to‑noise
ratio (SNR) of the stochastic gradient. Because
the effective step size \(\Delta_t = \alpha \cdot
\hat{m}_t / (\sqrt{\hat{v}_t}+\epsilon)\) is
proportional to this SNR, when the SNR is low
(e.g., near a minimum where gradient variance
dominates) the step size automatically shrinks,
providing an implicit annealing mechanism without
an explicit learning‑rate schedule.

6628106.c6

Unified Adaptive Gradient Method Framework

Adam: A Method for Stochastic Optimization

The paper synthesizes a conceptual framework that
positions Adam as a unifying adaptive optimizer.
It demonstrates that Adam subsumes the
sparse‑gradient handling of AdaGrad and the
non‑stationary adaptability of RMSProp, and it
formally relates Adam to AdaDelta, vSGD, and
natural‑gradient methods by expressing their
update rules within a common mathematical
formulation. This unified view clarifies how
different adaptive learning‑rate schemes can be
derived as special cases or limits of Adam’s
moment‑based updates.

229371222.c0

Iterative generative ProofWriter model for
faithful natural-language proof generation

ProofWriter: Generating Implications, Proofs, and
Abductive Statements over Natural Language

ProofWriter fine‑tunes a T5‑11B text‑to‑text
transformer to generate logical implications and
human‑readable proofs over natural‑language
theories by repeatedly applying a 1‑step
implication generator, appending each derived fact
to the context, and assembling deeper proofs from
these fragments. The iterative forward‑chaining
loop guarantees that every proof step corresponds
to a model decision, yielding proofs that are
faithful to the model’s internal reasoning.
Empirical evaluation on the RuleTaker,
Birds‑Electricity, and ParaRules datasets
demonstrates state‑of‑the‑art proof accuracy (+9 %
absolute over PRover) and robust generalisation to
reasoning depths unseen during training.

Prerequisite Edge Description

Description:
A large‑scale encoder‑decoder model pretrained on
massive text corpora and fine‑tuned for downstream
tasks via a unified text‑to‑text format.

Explanation:
ProofWriter builds directly on a fine‑tuned T5‑11B
model to generate implications and proofs.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A large‑scale encoder‑decoder model pretrained on
massive text corpora and fine‑tuned for downstream
tasks via a unified text‑to‑text format.

Explanation:
ProofWriter builds directly on a fine‑tuned T5‑11B
model to generate implications and proofs.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A synthetic benchmark of natural‑language facts,
rules, and queries where transformers have been shown
to emulate Datalog‑style deduction.

Explanation:
ProofWriter is trained and evaluated on RuleTaker
variants; the feasibility of neural logical deduction
on NL theories is established by this prior work.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A synthetic benchmark of natural‑language facts,
rules, and queries where transformers have been shown
to emulate Datalog‑style deduction.

Explanation:
ProofWriter is trained and evaluated on RuleTaker
variants; the feasibility of neural logical deduction
on NL theories is established by this prior work.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A system that assembles proofs by classifying proof
components and enforcing consistency with an ILP
solver, but whose proofs are post‑hoc and not
guaranteed to reflect model decisions.

Explanation:
PRover serves as the main prior art that motivates the
need for a faithful, generative proof‑generation
approach.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A system that assembles proofs by classifying proof
components and enforcing consistency with an ILP
solver, but whose proofs are post‑hoc and not
guaranteed to reflect model decisions.

Explanation:
PRover serves as the main prior art that motivates the
need for a faithful, generative proof‑generation
approach.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A system that assembles proofs by classifying proof
components and enforcing consistency with an ILP
solver, but whose proofs are post‑hoc and not
guaranteed to reflect model decisions.

Explanation:
PRover serves as the main prior art that motivates the
need for a faithful, generative proof‑generation
approach.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
The idea that a generated proof should correspond to
the model’s actual inference steps rather than being a
post‑hoc rationalisation.

Explanation:
ProofWriter’s iterative design is explicitly motivated
by the need for faithful proofs, as discussed in the
paper.

strength: strong; depth: 1

204838007.c9-1

Scaling model size up to 11 B parameters

Exploring the Limits of Transfer Learning with a
Unified Text-to-Text Transformer

The authors systematically increase the number of
parameters of the T5 encoder‑decoder Transformer
from the baseline 220 M to 770 M, 3 B and finally
11 B, keeping the architecture, pre‑training
objective, data, and training hyper‑parameters
constant. Each model is pre‑trained on the C4
corpus and fine‑tuned on a wide suite of NLP
benchmarks. The results show consistent
performance gains across tasks as model size
grows, confirming that larger models yield better
language understanding and generation
capabilities.

Prerequisite Edge Description

Description:
Design principles for scaling Transformer depth,
hidden size, feed‑forward dimension and attention
heads, as demonstrated in the BERT Large
configuration.

Explanation:
The larger T5 models (770 M, 3 B, 11 B) follow the
same scaling principles, requiring knowledge of how to
increase model dimensions while preserving training
stability.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
An adaptive learning‑rate optimizer with sublinear
memory cost, used for both pre‑training and
fine‑tuning of T5 models.

Explanation:
All scaling experiments use AdaFactor for
optimization, making its algorithm a prerequisite for
reproducing the results.

strength: strong; depth: 2

52967399.c6

Comprehensive ablation analysis of BERT pretraining objectives, model size, masking strategies, and training steps

BERT: Pre-training of Deep Bidirectional
Transformers for Language Understanding

The paper conducts a systematic set of ablation
experiments that (i) remove the Next Sentence
Prediction (NSP) task, (ii) replace the Masked
Language Model (MLM) objective with a
left‑to‑right language model, (iii) vary the depth
and width of the Transformer encoder, (iv) test
different token‑masking policies (MASK/SAME/RND),
and (v) change the number of pre‑training
optimization steps. These studies quantify how
each design choice impacts downstream performance
on sentence‑pair and token‑level benchmarks,
demonstrating that NSP is essential for
sentence‑pair tasks, bidirectional MLM outperforms
unidirectional LM, larger models yield gains even
on low‑resource tasks, and a mixed masking
strategy provides the best feature‑based results.
The findings are reported in Section 5 and
Appendix C.

4786918.c6

Adafactor optimizer: integrated low‑memory adaptive method

Adafactor: Adaptive Learning Rates with Sublinear
Memory Cost

Adafactor combines a factored approximation of the
second‑moment accumulator, removal of the momentum
term, update clipping, a time‑varying
second‑moment decay schedule, and relative
step‑size scaling into a single optimizer. The
algorithm matches Adam's performance on the
Transformer model for WMT 2014 English‑German
translation while requiring only O(n+m) auxiliary
memory for an n×m weight matrix. Full algorithmic
details are given in Algorithms 4‑6 and default
hyper‑parameters in Algorithm 6, and an
implementation is released in the Tensor2Tensor
library.

204838007.c11

Release of T5 model checkpoints, code, and C4
dataset

Exploring the Limits of Transfer Learning with a
Unified Text-to-Text Transformer

The authors publicly release pre‑trained T5 model
checkpoints for five model sizes (Small, Base,
Large, 3 B, 11 B), the full TensorFlow‑based
training and evaluation codebase, and the Colossal
Clean Crawled Corpus (C4) dataset. These artifacts
enable the research community to reproduce the
experiments, fine‑tune the models on downstream
tasks, and build new systems using the same
unified text‑to‑text framework. The release is
accompanied by documentation and integration with
TensorFlow Datasets for easy access.

Prerequisite Edge Description

Description:
A language‑independent subword tokenization library
that produces a shared 32 k wordpiece vocabulary for
both encoder and decoder.

Explanation:
Required for preprocessing the C4 corpus and for
tokenizing inputs/outputs of the released models.

strength: strong; depth: 2

52051958.c0

SentencePiece open‑source toolkit (Apache 2)

SentencePiece: A simple and language independent
subword tokenizer and detokenizer for Neural Text
Processing

SentencePiece is a publicly released C++/Python
library that provides command‑line tools
(spm_train, spm_encode, spm_decode) for
language‑independent subword tokenization and
detokenization. It supports both
byte‑pair‑encoding (BPE) and unigram
language‑model segmentation, performs Unicode NFKC
normalization, and manages a self‑contained model
file (Protocol Buffers) with vocabulary IDs. The
toolkit also offers C++, Python and TensorFlow
APIs for on‑the‑fly processing, enabling
end‑to‑end neural text pipelines without external
language‑specific preprocessors.

211126663.c1

Synthetic RuleTaker dataset suite

Transformers as Soft Reasoners over Language

The authors construct five large‑scale synthetic
benchmark datasets (D=0, D≤1, D≤2, D≤3, DMax),
each containing 100 k examples of randomly
generated rule‑based logical theories. For every
theory they perform exhaustive forward chaining to
obtain all derivable facts, label true statements,
generate false statements under a closed‑world
assumption, and render facts, rules, and queries
as templated English sentences. The datasets
systematically vary the maximum inference depth
(up to five steps) and include both negated and
non‑negated variants, providing a controlled
resource for training and evaluating neural models
on multi‑step logical reasoning over language.

211126663.c2

RuleTaker: RoBERTa‑large fine‑tuned as a soft theorem prover

Transformers as Soft Reasoners over Language

The authors fine‑tune a pretrained RoBERTa‑large
model (additionally fine‑tuned on the RACE
reading‑comprehension dataset) to predict the
truth value of a query given a natural‑language
context consisting of facts and
conjunctive‑implication rules. The resulting
model, called RuleTaker, emulates the input‑output
behavior of a formal theorem prover, operating
directly on language without any explicit symbolic
manipulation, and achieves near‑perfect accuracy
on synthetic rule‑reasoning benchmarks as well as
strong zero‑shot transfer to hand‑authored and
paraphrased rulebases.

Prerequisite Edge Description

Description:
A large‑scale transformer model pretrained on massive
text corpora using the RoBERTa training recipe,
providing rich contextual representations for
downstream fine‑tuning.

Explanation:
RuleTaker builds directly on RoBERTa‑large; without
the pretrained weights the model would lack the
linguistic knowledge needed to process the
natural‑language facts and rules.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
An additional fine‑tuning step on the RACE dataset,
which contains multiple‑choice reading‑comprehension
questions derived from examinations.

Explanation:
The authors report that this intermediate fine‑tuning
improves sensitivity to hyper‑parameters and overall
downstream performance before training on the
RuleTaker data.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
An additional fine‑tuning step on the RACE dataset,
which contains multiple‑choice reading‑comprehension
questions derived from examinations.

Explanation:
The authors report that this intermediate fine‑tuning
improves sensitivity to hyper‑parameters and overall
downstream performance before training on the
RuleTaker data.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Empirical findings that additional fine‑tuning on
task‑relevant data and careful hyper‑parameter choices
can boost transformer performance on downstream tasks.

Explanation:
The authors apply these techniques (e.g., fine‑tuning
on RACE) to improve RuleTaker’s accuracy, making them
a supporting methodological prerequisite.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Empirical findings that additional fine‑tuning on
task‑relevant data and careful hyper‑parameter choices
can boost transformer performance on downstream tasks.

Explanation:
The authors apply these techniques (e.g., fine‑tuning
on RACE) to improve RuleTaker’s accuracy, making them
a supporting methodological prerequisite.

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strength: strong; depth: 2

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task‑relevant data and careful hyper‑parameter choices
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Explanation:
The authors apply these techniques (e.g., fine‑tuning
on RACE) to improve RuleTaker’s accuracy, making them
a supporting methodological prerequisite.

strength: strong; depth: 2

198953378.c7

Open‑source implementation and pretrained checkpoints

RoBERTa: A Robustly Optimized BERT Pretraining
Approach

The authors release a full PyTorch/Fairseq
codebase that re‑implements BERT pretraining and
downstream fine‑tuning, together with pretrained
RoBERTa checkpoints for both BASE and LARGE
configurations. The released artifacts include
data preprocessing, dynamic masking, large‑batch
training, mixed‑precision support, and a
byte‑level BPE tokenizer, enabling reproducibility
of the RoBERTa recipe and allowing the community
to use state‑of‑the‑art models without retraining
from scratch.

6826032.c0

RACE Dataset and Benchmark for Machine Reading Comprehension

RACE: Large-scale ReAding Comprehension Dataset
From Examinations

The paper introduces RACE, a large‑scale English
reading‑comprehension resource containing 27,933
passages and 97,687 multiple‑choice questions
sourced from Chinese middle‑school and high‑school
English examinations. Each question provides four
human‑written answer options that are not
constrained to be text spans in the passage,
requiring deeper reasoning. The authors publish
predefined training, development, and test splits
for the middle‑school (RACE‑M) and high‑school
(RACE‑H) subsets and adopt simple accuracy as the
evaluation metric, thereby establishing a new
benchmark for the community.

6826032.c5

Open‑source Release of RACE Dataset and Baseline
Code

RACE: Large-scale ReAding Comprehension Dataset
From Examinations

The authors publicly release the full RACE
reading‑comprehension dataset (27,933 passages and
97,687 multiple‑choice questions) via a CMU
website and provide open‑source implementations of
three baseline models (Sliding Window, Stanford
Attentive Reader, Gated‑Attention Reader) on
GitHub. This release enables reproducible
benchmarking, comparative evaluation, and lowers
the entry barrier for future
machine‑reading‑comprehension research.

153312532.c0

Three‑step BERT fine‑tuning framework for text
classification

How to Fine-Tune BERT for Text Classification?

The paper introduces a systematic three‑stage
pipeline to adapt a pretrained BERT model to any
text classification task. First, BERT is further
pre‑trained on the target‑task training set or on
additional in‑domain data using the original
masked language modeling and
next‑sentence‑prediction objectives. Second, an
optional multi‑task fine‑tuning stage jointly
trains the shared BERT encoder on several related
classification tasks while keeping task‑specific
output heads. Third, the target task is fine‑tuned
with a layer‑wise decreasing learning‑rate
schedule and a long‑text preprocessing strategy,
yielding state‑of‑the‑art results on eight
benchmark datasets.

153312532.c2

Layer-wise learning-rate decay schedule to mitigate catastrophic forgetting in BERT fine-tuning

How to Fine-Tune BERT for Text Classification?

The paper introduces a learning‑rate schedule for
BERT fine‑tuning that assigns a base learning rate
to the top Transformer layer and geometrically
decays it for lower layers using a factor \(\xi
\le 1\). Setting \(\xi = 0.95\) with a base rate
of 2e‑5 yields a lower error on the IMDb sentiment
classification task by reducing catastrophic
forgetting. The schedule is evaluated empirically
on IMDb and analyzed in terms of its effect on
knowledge retention across BERT layers.

153312532.c4

Task‑specific, in‑domain, and cross‑domain further pre‑training of BERT

How to Fine-Tune BERT for Text Classification?

The authors augment the original BERT model with
an additional unsupervised pre‑training phase that
re‑applies the masked language modeling (MLM) and
next‑sentence prediction (NSP) objectives on (i)
the target task’s own training corpus
(within‑task), (ii) a collection of corpora
sharing the same semantic domain (in‑domain), and
(iii) the union of all seven classification
corpora (cross‑domain). This further pre‑training
is inserted as the first stage of their three‑step
BERT fine‑tuning pipeline and is evaluated on
eight public text‑classification benchmarks, where
it consistently reduces error rates relative to
vanilla BERT‑base, with the largest gains observed
for in‑domain pre‑training on small datasets such
as TREC.

153312532.c5

Multi‑task fine‑tuning of BERT for text
classification

How to Fine-Tune BERT for Text Classification?

The authors jointly fine‑tune a shared BERT‑base
encoder on four English text‑classification
datasets (IMDb, Yelp‑P, AG's News, DBPedia) using
a multi‑task learning setup where each task has a
private classification head. After the joint
training phase, they perform an additional
low‑learning‑rate fine‑tuning step on each
individual task. Experiments show modest accuracy
gains over single‑task fine‑tuning and reveal that
the benefit of multi‑task fine‑tuning diminishes
when the model has already been subjected to
cross‑domain further pre‑training.

153312532.c6

Few‑shot learning performance of (further pre‑trained) BERT for text classification

How to Fine-Tune BERT for Text Classification?

The authors evaluate the pretrained BERT‑base
model and its variants that undergo additional
in‑domain pre‑training on the IMDb sentiment
classification task when only a very small
fraction (0.4 % to 5 %) of the original training
data is available. By fine‑tuning BERT on these
limited subsets they observe a reduction in error
rate from 17.26 % to 9.23 % with the standard
model, and further gains when the model has been
further pre‑trained on in‑domain data,
demonstrating BERT's suitability for low‑resource
text classification scenarios.

222141025.c0-1

PROVER Multi-Module Transformer Architecture for
Joint QA and Proof Generation

PROVER: Proof Generation for Interpretable
Reasoning over Rules

PROVER is a neural system built on a shared
RoBERTa‑large encoder that simultaneously performs
binary question answering over natural‑language
rule bases and constructs explicit proof graphs.
It consists of three parallel modules: a QA
classifier that predicts the answer, a Node
classifier that selects the subset of facts,
rules, and a Negation‑As‑Failure (NAF) node
required for the proof, and an Edge classifier
that predicts directed edges between the selected
nodes to form a proof graph. The three modules are
trained jointly with a combined cross‑entropy
loss, enabling shared representations useful for
both answer prediction and structured proof
construction.

Prerequisite Edge Description

Description:
A large‑scale transformer language model pre‑trained
on massive text corpora, providing contextual token
embeddings and a global [CLS] representation.

Explanation:
PROVER uses RoBERTa‑large as the backbone encoder for
all three modules.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A large‑scale transformer language model pre‑trained
on massive text corpora, providing contextual token
embeddings and a global [CLS] representation.

Explanation:
PROVER uses RoBERTa‑large as the backbone encoder for
all three modules.

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Prerequisite Edge Description

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A large‑scale transformer language model pre‑trained
on massive text corpora, providing contextual token
embeddings and a global [CLS] representation.

Explanation:
PROVER uses RoBERTa‑large as the backbone encoder for
all three modules.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
The self‑attention based neural network architecture
that processes sequences in parallel and captures
long‑range dependencies.

Explanation:
PROVER’s encoder and the design of its modules rely on
the transformer model paradigm.

strength: strong; depth: 2

198953378.c4

RoBERTa pretraining recipe and state‑of‑the‑art
benchmark results

RoBERTa: A Robustly Optimized BERT Pretraining
Approach

The authors introduce RoBERTa, a robustly
optimized BERT pretraining configuration that
combines (i) dynamic masking, (ii) removal of the
next‑sentence prediction (NSP) objective, (iii)
large mini‑batch training with appropriate
learning‑rate scaling, (iv) training on
full‑length sequences throughout, and (v) a
byte‑level BPE vocabulary. Using this recipe they
train BERT‑BASE and BERT‑LARGE models for up to
500 K steps on up to 160 GB of public text
(including the CC‑NEWS corpus). The resulting
models achieve new best scores on GLUE, SQuAD
1.1/2.0 and RACE, surpassing prior work such as
XLNet without multi‑task finetuning or extra data.

198953378.c1

Discovery of BERT under‑training and four simple training improvements (RoBERTa recipe)

RoBERTa: A Robustly Optimized BERT Pretraining
Approach

By replicating BERT pretraining, the authors show
that the original BERT model was trained for
insufficient steps and that its performance can be
substantially improved by (i) training for many
more updates with larger mini‑batches, (ii)
removing the next‑sentence prediction (NSP)
objective, (iii) feeding longer input sequences
(full‑sentence inputs), and (iv) using dynamic
masking instead of a static mask. These four
modifications together define the RoBERTa training
recipe, which achieves state‑of‑the‑art results on
GLUE, SQuAD and RACE without extra data or
multi‑task finetuning.

222141025.c0-2

Constrained Joint Training Regime with Edge Masking and Global Semantic Constraints

PROVER: Proof Generation for Interpretable
Reasoning over Rules

PROVER jointly predicts binary labels for node
presence and edge existence while masking out
impossible edges (self‑loops, fact‑to‑fact,
rule‑to‑fact, and edges involving absent nodes)
during training, and combines the QA, Node, and
Edge binary cross‑entropy losses into a single
objective. At inference, an Integer Linear
Programming (ILP) solver selects a subset of edges
that maximizes the summed edge probabilities while
satisfying semantic edge constraints and a
max‑flow connectivity constraint, guaranteeing
that only valid, connected proof graphs are
produced.

Prerequisite Edge Description

Description:
A large pre‑trained language model that provides
contextual token embeddings for the shared encoder
used by the QA, Node, and Edge modules.

Explanation:
PROVER relies on RoBERTa to obtain rich
representations of facts, rules, and the question,
which are essential for predicting node and edge
labels.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A large pre‑trained language model that provides
contextual token embeddings for the shared encoder
used by the QA, Node, and Edge modules.

Explanation:
PROVER relies on RoBERTa to obtain rich
representations of facts, rules, and the question,
which are essential for predicting node and edge
labels.

strength: strong; depth: 2

222141025.c1

Graph‑Based Proof Representation with Global
Constraints

PROVER: Proof Generation for Interpretable
Reasoning over Rules

The paper formalizes proofs for natural‑language
rule reasoning as directed graphs whose nodes are
facts, rules, or a special Negation‑As‑Failure
(NAF) node and whose edges encode permissible
inference directions (fact→rule or rule→rule).
Three proof types are defined (successful with
NAF, successful without NAF, failed) and during
training impossible edges (self‑loops,
fact‑to‑fact, rule‑to‑fact) are masked to simplify
learning. At inference time an Integer Linear
Program selects a subset of edges that maximizes
the model's edge scores while enforcing global
constraints such as graph connectivity and the
semantic edge restrictions, guaranteeing that the
generated proof is valid.

Prerequisite Edge Description

Description:
RuleTakers fine‑tunes a RoBERTa model to answer binary
questions over rule‑bases expressed in natural
language, treating transformers as "soft theorem
provers".

Explanation:
Provides the task definition, data format, and
baseline QA model that PROVER extends with proof‑graph
generation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
RuleTakers fine‑tunes a RoBERTa model to answer binary
questions over rule‑bases expressed in natural
language, treating transformers as "soft theorem
provers".

Explanation:
Provides the task definition, data format, and
baseline QA model that PROVER extends with proof‑graph
generation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
RoBERTa‑large is a transformer model pre‑trained on
large corpora with optimized training
hyper‑parameters, yielding strong contextual
representations for downstream tasks.

Explanation:
Serves as the shared encoder whose token embeddings
are pooled to represent facts, rules, and the NAF node
for the proof‑graph construction.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
RoBERTa‑large is a transformer model pre‑trained on
large corpora with optimized training
hyper‑parameters, yielding strong contextual
representations for downstream tasks.

Explanation:
Serves as the shared encoder whose token embeddings
are pooled to represent facts, rules, and the NAF node
for the proof‑graph construction.

strength: strong; depth: 2

211126663.c0

RuleTaker soft theorem‑proving task formulation

Transformers as Soft Reasoners over Language

The paper defines a new reasoning task in which a
transformer model receives a context consisting of
natural‑language facts and conjunctive‑implication
rules and must answer a true/false query that may
require multi‑step deduction. By expressing the
knowledge base directly in English rather than a
formal logical language, the task reframes logical
inference as a "soft theorem proving" problem that
can be learned end‑to‑end. This formulation
isolates the I/O behavior of a deductive reasoner
and enables systematic evaluation of neural models
on controlled inference depths.

218487535.c0

Module-wise Faithfulness Framework for Neural
Module Networks

Obtaining Faithful Interpretations from
Compositional Neural Networks

The paper introduces a systematic framework to
evaluate whether each neural module in a Neural
Module Network (NMN) performs its intended atomic
reasoning operation. It defines the notion of
module-wise faithfulness and proposes quantitative
metrics: precision/recall/F1 for visual NMNs based
on aligned bounding-box annotations, and
cross-entropy for textual NMNs based on gold span
annotations. The framework enables rigorous,
reproducible assessment of interpretability beyond
ad-hoc qualitative inspection.

Prerequisite Edge Description

Description:
A class of models that parse a natural-language
utterance into an executable program composed of
learnable modules, each intended to perform an atomic
reasoning operation, and execute the program against a
visual or textual context.

Explanation:
The faithfulness framework is defined with respect to
the internal modules and program structure of NMNs, so
understanding NMNs is essential.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A class of models that parse a natural-language
utterance into an executable program composed of
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visual or textual context.

Explanation:
The faithfulness framework is defined with respect to
the internal modules and program structure of NMNs, so
understanding NMNs is essential.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A dataset of paired images and compositional
natural-language statements requiring multi-step
visual reasoning; the task is to predict a Boolean
truth value for each statement.

Explanation:
The visual component of the faithfulness framework is
evaluated on NLVR2, requiring knowledge of its task
formulation and data format.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A dataset of paired images and compositional
natural-language statements requiring multi-step
visual reasoning; the task is to predict a Boolean
truth value for each statement.

Explanation:
The visual component of the faithfulness framework is
evaluated on NLVR2, requiring knowledge of its task
formulation and data format.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A dataset of paragraphs, questions, and answers that
require discrete reasoning (e.g., counting,
arithmetic, date comparison) over the text; answers
are spans, numbers, or dates.

Explanation:
The textual component of the faithfulness framework is
demonstrated on DROP, so familiarity with its task and
evaluation is required.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A dataset of paragraphs, questions, and answers that
require discrete reasoning (e.g., counting,
arithmetic, date comparison) over the text; answers
are spans, numbers, or dates.

Explanation:
The textual component of the faithfulness framework is
demonstrated on DROP, so familiarity with its task and
evaluation is required.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior work that defines and studies the notion of
faithful explanations for neural networks, emphasizing
that model internals should align with
human-interpretable reasoning steps.

Explanation:
The framework builds on the concept of faithfulness
introduced in earlier work, motivating the need for
module-wise evaluation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior work that defines and studies the notion of
faithful explanations for neural networks, emphasizing
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Explanation:
The framework builds on the concept of faithfulness
introduced in earlier work, motivating the need for
module-wise evaluation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior work that defines and studies the notion of
faithful explanations for neural networks, emphasizing
that model internals should align with
human-interpretable reasoning steps.

Explanation:
The framework builds on the concept of faithfulness
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module-wise evaluation.

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Prerequisite Edge Description

Description:
A standard metric that measures the overlap between
two bounding boxes; used to align model-proposed boxes
with human-annotated boxes for computing precision and
recall.

Explanation:
The visual faithfulness metric relies on IoU to decide
whether a predicted box matches a gold box.

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with human-annotated boxes for computing precision and
recall.

Explanation:
The visual faithfulness metric relies on IoU to decide
whether a predicted box matches a gold box.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A deep bidirectional transformer model pre-trained on
large corpora, providing contextual token embeddings
for the Text-NMN.

Explanation:
Text-NMN uses BERT to encode questions and passages;
while not central to the faithfulness definition, it
underlies the module inputs.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A transformer-based model that jointly encodes
language and visual inputs, used as the backbone for
the Visual-NMN.

Explanation:
Visual-NMN builds on LXMERT representations;
understanding this architecture is required to
interpret the visual module outputs.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Crowd-sourced decompositions of NLVR2 questions into
executable programs with utterance attention,
providing the ground-truth program structure used for
evaluation.

Explanation:
The module-wise faithfulness evaluation requires gold
programs to isolate module behavior from parser
errors.

strength: strong; depth: 2

3130692.c0

Dynamic Neural Module Network (D‑NMN) architecture

Learning to Compose Neural Networks for Question
Answering

The paper proposes a Dynamic Neural Module Network
that jointly learns a set of composable neural
modules and a layout predictor which assembles a
question‑specific deep network on the fly. The
system is trained end‑to‑end from (world,
question, answer) triples without any explicit
supervision of the network layouts, thereby
unifying visual question answering and structured
knowledge‑base QA under a single differentiable
framework. The architecture combines
attention‑based modules, a dependency‑parse‑driven
layout candidate generator, and a REINFORCE‑based
policy to select layouts, enabling expressive
instance‑specific computation.

3130692.c1

Joint learning algorithm for modules and layouts

Learning to Compose Neural Networks for Question
Answering

The paper introduces a training procedure that
simultaneously learns the parameters of a set of
neural modules and the discrete layout predictor
that assembles them into question-specific
networks. Module parameters are updated with
standard back‑propagation, while the layout
predictor is trained with a REINFORCE
policy‑gradient where the reward is the negative
log‑likelihood of the correct answer. This enables
end‑to‑end learning from only (world, question,
answer) triples without any layout supervision.

3130692.c2

Neural module inventory for visual and structured
domains

Learning to Compose Neural Networks for Question
Answering

The paper defines a set of six composable neural
modules—lookup, find, relate, and, describe, and
exists—each with explicit input and output type
constraints (Attention or Labels). These modules
extend the standard soft‑attention mechanism to
operate both on convolutional image feature maps
and on vector embeddings of entities from
knowledge bases, providing differentiable
analogues of logical operations such as set
intersection and existential quantification.
Modules share global parameters across instances
while allowing lexical specialization via
per‑module parameter arguments, enabling reuse and
efficient learning.

3130692.c3

Layout prediction via dependency‑parse candidates and LSTM‑MLP scoring

Learning to Compose Neural Networks for Question
Answering

The paper introduces a method that converts a
question into a small set of candidate network
layouts by first parsing the sentence with a
Stanford dependency parser, then mapping nouns,
verbs, and prepositional phrases to predefined
module fragments. Each candidate layout is scored
by concatenating an LSTM encoding of the whole
question with a feature vector describing the
layout and feeding the result through a multilayer
perceptron; the scores are normalized to a
distribution p(z|x). During training the layout is
sampled and the scoring network is updated with a
REINFORCE policy‑gradient signal, allowing the
model to learn layout selection without explicit
layout supervision.

18682.c0

End-to-End Module Networks (N2NMN) architecture

Learning to Reason: End-to-End Module Networks for
Visual Question Answering

The paper introduces End-to-End Module Networks
(N2NMN), a visual question answering system that
dynamically assembles a question‑specific neural
network from a library of co‑attentive modules. A
sequence‑to‑sequence controller with attention
predicts a layout expression in reverse‑Polish
notation directly from the raw question, and each
module receives a soft‑attention‑derived textual
vector, eliminating hand‑crafted word assignments.
Training proceeds in two stages: behavioral
cloning from expert layouts provides a strong
initialization, followed by REINFORCE‑style
policy‑gradient fine‑tuning of the layout policy
and module parameters. The resulting model
achieves state‑of‑the‑art performance on CLEVR and
competitive results on VQA and SHAPES.

18682.c2

Co‑attentive neural modules for visual reasoning

Learning to Reason: End-to-End Module Networks for
Visual Question Answering

The paper defines a library of differentiable
neural modules (e.g., find, filter, relocate, and,
or, describe, compare, count, exist) that operate
on zero, one, or two image‑attention maps together
with visual feature maps and a
soft‑attention‑derived textual vector. Each module
produces either a refined attention map or a
distribution over answer candidates, providing
reusable functional primitives for compositional
visual question answering. The modules are
designed to be assembled dynamically according to
a layout predicted from the question, enabling
end‑to‑end training of both module parameters and
the layout policy.

202565945.c0

Self-assembling Neural Modular Network for Text-based Multi-hop QA

Self-Assembling Modular Networks for Interpretable
Multi-Hop Reasoning

The paper introduces the first Neural Modular
Network (NMN) that self‑assembles a reasoning
architecture for unstructured text. A recurrent
controller reads a HotpotQA question, generates
soft sub‑question vectors, and predicts a soft
layout over four language‑oriented modules (Find,
Relocate, Compare, NoOp). The selected modules are
executed on a differentiable stack that stores
intermediate attention maps, enabling end‑to‑end
gradient‑based training without reinforcement
learning. This design provides flexible,
compositional reasoning over long contexts
containing multiple evidence pieces and yields
state‑of‑the‑art performance on HotpotQA and its
adversarial variant.

202565945.c1

Four language‑oriented NMN modules for textual reasoning

Self-Assembling Modular Networks for Interpretable
Multi-Hop Reasoning

The paper introduces four novel neural
modules—Find, Relocate, Compare, and
NoOp—specifically designed for multi‑hop question
answering over unstructured text. Find performs
bidirectional attention between the question and
the context to produce an attention map over
answer spans. Relocate consumes the attention from
a previous step, extracts a bridge‑entity
representation, and searches the context for the
next answer. Compare aggregates two attention maps
and evaluates relational properties between the
extracted entities. NoOp passes the current
network state unchanged, allowing the controller
to terminate reasoning. These modules replace the
image‑centric operations used in prior NMNs and
enable end‑to‑end differentiable multi‑step
reasoning for HotpotQA.

202565945.c2

Controller RNN with soft layout prediction and sub‑question attention

Self-Assembling Modular Networks for Interpretable
Multi-Hop Reasoning

A recurrent controller reads the encoded HotpotQA
question, computes a soft attention distribution
over its words to produce a sub‑question vector,
and at each reasoning step predicts a probability
distribution over the four NMN modules (Find,
Relocate, Compare, NoOp). The soft layout weights
the outputs of all modules, enabling
differentiable program execution without
reinforcement learning, while the sub‑question
vector updates the controller hidden state for the
next step, allowing fully end‑to‑end training of
the modular reasoning network.

53178856.c1

NLVR2: large‑scale natural‑language visual reasoning dataset and benchmark

A Corpus for Reasoning About Natural Language
Grounded in Photographs

The paper introduces NLVR2, a publicly released
corpus of 107,292 examples consisting of 29,680
distinct English captions each paired with two
real‑world photographs (127,502 unique images).
Each caption appears with multiple image‑pair
instances that are labeled True or False in a
balanced fashion, defining a binary verification
task that requires joint reasoning over language
and two images. Fixed train/dev/public‑test
splits, a consistency metric, and baseline
evaluations are provided, establishing NLVR2 as a
new benchmark for compositional visual reasoning.

53178856.c8

Public release of NLVR2 dataset, collection pipeline, and evaluation code

A Corpus for Reasoning About Natural Language
Grounded in Photographs

The authors make publicly available the full NLVR2
corpus (train, development, and public‑test
splits), the end‑to‑end image‑collection and
caption‑writing pipelines, and the source code for
baseline and adapted visual‑reasoning models
together with evaluation scripts at
http://lil.nlp.cornell.edu/nlvr/. This release
enables reproducibility, benchmarking, and further
research on compositional visual reasoning with
real photographs.

67855846.c1

Creation of the DROP dataset (96,567 QA pairs with numbers, spans, dates)

DROP: A Reading Comprehension Benchmark Requiring
Discrete Reasoning Over Paragraphs

The authors construct a large‑scale English
reading‑comprehension benchmark consisting of
96,567 question‑answer pairs drawn from Wikipedia
passages. Answers are limited to text spans,
numeric values, or dates, and the questions are
designed to require discrete reasoning such as
addition, subtraction, counting, sorting, and
coreference resolution. An adversarial
crowdsourcing pipeline forces workers to produce
questions that a baseline BiDAF model cannot
answer, ensuring high difficulty. The dataset is
split by passage into training, development, and
test sets and released publicly as a benchmark for
evaluating discrete reasoning over paragraphs.

67855846.c6

Public release of the DROP benchmark, baseline implementations, and evaluation leaderboard

DROP: A Reading Comprehension Benchmark Requiring
Discrete Reasoning Over Paragraphs

The authors publicly release the DROP dataset
together with reference implementations of
baseline reading‑comprehension models (BiDAF,
QANet, QANet+ELMo, BERT) and the NAQANet model,
and provide a hidden‑test leaderboard hosted at
https://allennlp.org/drop. This release supplies
the research community with the data, code, and
evaluation infrastructure needed to develop and
compare systems that perform discrete reasoning
over paragraphs.

7053611.c0

Input‑gradient explanations match LIME fidelity and are computationally efficient

Right for the Right Reasons: Training
Differentiable Models by Constraining their
Explanations

The paper demonstrates that the gradient of a
differentiable model’s output with respect to its
inputs yields local, feature‑level explanations
that closely resemble those produced by the
model‑agnostic LIME method across toy, text, and
image datasets. By visualizing the magnitude of
input gradients, the same salient pixels or words
identified by LIME are recovered, while the
gradient can be obtained with a single backward
pass, making it orders of magnitude faster than
LIME’s perturb‑and‑fit procedure. This establishes
input‑gradient explanations as a faithful and
scalable alternative for interpreting neural
networks.

7053611.c1

Gradient‑based explanation regularization (RRR)

Right for the Right Reasons: Training
Differentiable Models by Constraining their
Explanations

The authors introduce a differentiable loss term
that augments the standard cross‑entropy and
weight‑decay objectives with a penalty on the
magnitude of input‑gradient components at feature
dimensions marked as irrelevant by a binary
annotation matrix A. By shrinking gradients in
these regions, training steers the model toward
decision boundaries that respect domain knowledge
while preserving classification accuracy. The
regularizer is compatible with any gradient‑based
optimizer and can be applied to arbitrary
differentiable models.

7053611.c3-1

Explanation‑regularized models retain accuracy
under distribution shift

Right for the Right Reasons: Training
Differentiable Models by Constraining their
Explanations

The authors augment the standard cross‑entropy
loss with a penalty on input‑gradient magnitudes
for features marked as irrelevant, training neural
classifiers that are constrained to rely on
domain‑relevant cues. Across four domains—a
synthetic 5×5×3 color dataset, a composite
Iris‑Cancer tabular set, the 20 Newsgroups text
classification task, and a Decoy MNIST image
set—they show that models trained with this
gradient‑based explanation regularization maintain
or recover baseline test accuracy when the
training‑test distribution differs (e.g., when
confounding features are removed). This
demonstrates that constraining explanations
improves robustness to covariate shift caused by
spurious correlations.

7053611.c4

Explanation regularization improves data
efficiency

Right for the Right Reasons: Training
Differentiable Models by Constraining their
Explanations

The authors empirically demonstrate on a synthetic
5×5×3 RGB toy color dataset that adding a
gradient‑based explanation penalty (RRR) or the
unsupervised find‑another‑explanation penalty
enables a model to achieve a target accuracy with
far fewer labeled training examples than a
baseline model without such regularization. By
varying the number of training samples and
measuring test accuracy, they show that
constraining explanations to align with domain
knowledge reduces the sample complexity of
learning. The analysis also compares the two
regularization strategies and discusses their
impact on data efficiency.

7053611.c5

Open‑source implementation of Right‑for‑the‑Right‑Reasons (RRR) training

Right for the Right Reasons: Training
Differentiable Models by Constraining their
Explanations

The authors release a publicly available software
package (https://github.com/dtak/rrr) that
implements the full RRR pipeline: generation of
input‑gradient explanations, the
explanation‑regularization loss that penalises
gradients on annotated irrelevant features, and
the find‑another‑explanation iterative algorithm
for discovering diverse classifiers. The code is
built on the autograd automatic‑differentiation
library and uses the Adam optimizer to train
two‑hidden‑layer multilayer perceptrons,
reproducing all experiments reported in the paper
and enabling other researchers to apply the method
to new models and datasets.

67855860.c0

Large‑scale empirical benchmark of attention as explanation

Attention is not Explanation

The paper conducts a systematic, large‑scale
empirical study of standard attention mechanisms
across nine NLP tasks (binary sentiment, IMDB, ADR
tweets, 20 News, AG News, MIMIC ICD‑9, CNN QA,
bAbI, SNLI) and three encoder families (Bi‑LSTM,
CNN, average‑embedding). For each dataset‑encoder
pair it quantifies (i) the Kendall‑τ correlation
between learned attention weights and two
alternative feature‑importance signals (input
gradients and leave‑one‑out erasure) and (ii) the
effect of altering attention distributions on
model predictions using total variation distance
and Jensen‑Shannon divergence, thereby providing
the first comprehensive benchmark of the
explanatory value of attention.

67855860.c1

Weak and inconsistent correlation between
attention weights and feature‑importance measures

Attention is not Explanation

The authors compute Kendall‑τ rank correlation
between attention scores generated by standard
additive (Bahdanau) and scaled‑dot‑product
(Vaswani) attention mechanisms and two established
feature‑importance baselines: input‑gradient
attributions and leave‑one‑out (erasure) scores.
Across nine NLP tasks using a Bi‑LSTM encoder, the
average τ never exceeds 0.36 and is often
statistically insignificant, whereas a simple
average‑embedding encoder yields substantially
higher τ (~0.6). This analysis demonstrates that,
for contextual encoders, attention weights are not
reliable proxies for the inputs that most drive
model predictions.

67855860.c2

Random permutation of attention weights often leaves predictions unchanged

Attention is not Explanation

The authors randomly permute the attention weight
vector produced by standard additive or scaled
dot‑product attention for each test instance and
measure the resulting change in model output using
Total Variation Distance (TVD). Across nine NLP
tasks (sentiment classification, QA, NLI) and
three encoder families (BiLSTM, CNN,
average‑embedding) they find many instances where
even large attention weights can be shuffled with
only negligible impact on the prediction,
indicating that the original attention heatmap is
not uniquely tied to the output.

67855860.c3

Adversarial attention optimization for counterfactual explanations

Attention is not Explanation

The paper formulates a constrained optimization
problem that seeks attention weight vectors
maximally dissimilar (high Jensen‑Shannon
divergence) from the model‑learned attention while
keeping the model output within a small
total‑variation distance (≤ 0.01 for
classification, ≤ 0.05 for QA). The objective is
solved with the Adam optimizer and includes a
diversity term to obtain multiple, diverse
adversarial attention distributions per instance.
Empirical experiments across nine NLP tasks show
that many distinct attention heatmaps can be
generated that leave the prediction virtually
unchanged, demonstrating that standard attention
weights are not unique explanations for model
outputs.

67855860.c4

Open‑source codebase and interactive visualisation
suite

Attention is not Explanation

The authors release a fully reproducible
implementation of all models, datasets, and
analysis pipelines used in the study, hosted at
https://github.com/successar/AttentionExplanation.
In addition, they provide an online interactive
interface that lets users explore attention
heatmaps, correlation histograms, and adversarial
examples for each dataset and encoder
configuration. This resource enables other
researchers to replicate the experiments, extend
the analysis, and benchmark future work on
attention as explanation.

67855860.c5

Position statement cautioning against treating attention as explanation

Attention is not Explanation

The authors assert that standard attention modules
in neural NLP models should not be interpreted as
providing meaningful explanations of model
predictions. This stance is grounded in extensive
empirical evidence showing (i) weak and
inconsistent correlation between attention weights
and established feature‑importance measures
(gradient saliency and leave‑one‑out erasure) and
(ii) the existence of many counterfactual
attention distributions that leave predictions
virtually unchanged. Consequently, they advise
practitioners to avoid presenting attention
heatmaps as definitive rationales and to consider
alternative interpretability techniques.

215416110.c0

Taxonomy of Faithfulness Evaluation Based on Three Core Assumptions

Towards Faithfully Interpretable NLP Systems: How
Should We Define and Evaluate Faithfulness?

The paper surveys existing faithfulness evaluation
methods for NLP models and organizes them around
three explicit assumptions: (1) the Model
Assumption – identical predictions imply identical
reasoning; (2) the Prediction Assumption – similar
inputs should lead to similar reasoning; and (3)
the Linearity Assumption – input parts contribute
independently to the decision. By making these
assumptions explicit, the authors provide a
unified conceptual taxonomy that clarifies how
disparate evaluation techniques are related and
what notion of faithfulness they implicitly adopt.

215416110.c2

Critique of Binary Faithfulness and Proposal of
Graded Faithfulness Criteria

Towards Faithfully Interpretable NLP Systems: How
Should We Define and Evaluate Faithfulness?

The paper argues that treating faithfulness of NLP
explanations as a binary property is unrealistic
because explanations are approximations of highly
non‑linear model reasoning and therefore cannot be
perfectly faithful for all inputs. It shows that
existing binary evaluations mainly serve to
falsify methods via counter‑examples and that such
a strict view hinders practical progress. To
address this, the authors propose a graded,
"grayscale" notion of faithfulness that quantifies
the degree to which an explanation aligns with the
model’s reasoning, enabling nuanced assessment
across models, tasks, and input regions.

215416110.c3

Explicit Distinction Between Faithfulness and Plausibility

Towards Faithfully Interpretable NLP Systems: How
Should We Define and Evaluate Faithfulness?

The authors highlight that the interpretability
literature often conflates faithfulness (the
degree to which an explanation mirrors the model's
actual reasoning) with plausibility (the degree to
which an explanation appears convincing to
humans). They analyze prior work to show how this
conflation can lead to harmful outcomes in
high‑stakes applications and argue for a clear,
separate treatment of the two criteria in both
research and practice.

215416110.c4

Two-Axis Graded Faithfulness Evaluation Framework

Towards Faithfully Interpretable NLP Systems: How
Should We Define and Evaluate Faithfulness?

The paper proposes a future research agenda that
evaluates explanation faithfulness along two
complementary dimensions: (1) across models and
tasks, measuring the degree of faithfulness for
each specific model‑task pair, and (2) across the
input space, measuring faithfulness locally for
neighborhoods of similar inputs or individual
instances. This two‑axis conceptual framework
replaces the prevailing binary notion of
faithfulness with a nuanced, context‑dependent
assessment that can capture partial but useful
alignment between explanations and model
reasoning. The authors also outline procedural
directions for operationalising these axes in
empirical studies, encouraging the community to
report graded faithfulness scores rather than a
simple faithful/unfaithful label.

10328909.c2

Multi-task loss for joint objectness
classification and bounding-box regression

Faster R-CNN: Towards Real-Time Object Detection
with Region Proposal Networks

The paper defines a unified loss function for
training Region Proposal Networks that
simultaneously optimizes a binary softmax
classification loss (object vs. background) and a
smooth‑L1 regression loss for bounding‑box
offsets. Positive anchors are defined by IoU > 0.7
(or highest overlap) and the regression term is
activated only for these anchors; the two terms
are balanced by a weighting factor λ = 10. This
multi‑task loss enables end‑to‑end learning of
high‑quality, class‑agnostic proposals in a single
training pass.

10328909.c7

Analysis of proposal recall versus detection mAP

Faster R-CNN: Towards Real-Time Object Detection
with Region Proposal Networks

The paper empirically evaluates how the
recall‑to‑IoU curves of three region proposal
methods (RPN, Selective Search, EdgeBoxes) relate
to the final object detection mean Average
Precision (mAP) obtained with Fast R‑CNN. By
measuring recall at multiple IoU thresholds and
comparing these curves to detection mAP, the
authors demonstrate that very high IoU recall
(>0.8) is not necessary for strong detection
performance, explaining why the RPN maintains
accuracy with only 300 proposals.

201103729.c0

LXMERT multi‑encoder cross‑modality architecture

LXMERT: Learning Cross-Modality Encoder
Representations from Transformers

The paper introduces LXMERT, a Transformer‑based
vision‑and‑language model that separates
processing into three encoders: (1) an
object‑relationship encoder that consumes a
sequence of region‑of‑interest (RoI) visual
features, (2) a language encoder that processes
WordPiece‑tokenized sentences, and (3) a
cross‑modality encoder that repeatedly applies
bi‑directional cross‑attention followed by
self‑attention to fuse visual and textual streams.
The architecture also defines position‑aware
object embeddings that combine projected RoI
features with learned bounding‑box embeddings,
enabling joint reasoning over visual concepts and
linguistic semantics.

211003735.c3

BREAK Dataset of 83,978 Question–QDMR Pairs

Break It Down: A Question Understanding Benchmark

The authors construct BREAK, a large‑scale dataset
comprising 83,978 natural‑language questions
sampled from ten existing QA benchmarks across
three modalities (structured databases, text
passages, and images) together with high‑quality
Question Decomposition Meaning Representations
(QDMRs) annotated via crowdsourcing. The dataset
is split into train, development, and test sets
that respect original context boundaries, and is
released publicly along with the annotation
interface, validation tools, and code. BREAK
serves both as a resource for training QDMR
parsers and as a benchmark for evaluating
downstream multi‑hop QA and semantic parsing
systems.

229371222.c1

All‑At‑Once proof generation baseline for natural‑language reasoning

ProofWriter: Generating Implications, Proofs, and
Abductive Statements over Natural Language

The paper introduces a single‑pass generative
baseline that fine‑tunes a pretrained T5
text‑to‑text model to map a natural‑language
theory and a query directly to an answer
(True/False/Unknown) together with a full
linearized proof and all intermediate conclusions.
The model is trained on the same data as the
iterative ProofWriter system but learns to emit
the entire proof in one generation step, enabling
direct comparison of proof quality, faithfulness,
and depth generalisation. The authors also propose
a verification procedure that re‑queries the same
model in QA mode for each intermediate step of the
generated proof to assess whether the proof
fragments are actually supported by the model.

Prerequisite Edge Description

Description:
A large sequence‑to‑sequence transformer model
pretrained on massive corpora and fine‑tuned for
downstream tasks via a unified text‑to‑text interface.

Explanation:
The All‑At‑Once baseline fine‑tunes T5 to generate
answers and full proofs in a single pass.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Synthetic datasets of English facts and rules
generated from Datalog programs, together with queries
and gold proofs, used to train and evaluate neural
logical reasoning.

Explanation:
Provides the training and test data on which the
All‑At‑Once model is trained and evaluated.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A prior system that classifies which facts, rules, and
connections belong in a proof graph and assembles them
using an ILP solver.

Explanation:
Serves as the main prior baseline that the All‑At‑Once
approach seeks to improve upon and compare against.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A prior system that classifies which facts, rules, and
connections belong in a proof graph and assembles them
using an ILP solver.

Explanation:
Serves as the main prior baseline that the All‑At‑Once
approach seeks to improve upon and compare against.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A prior system that classifies which facts, rules, and
connections belong in a proof graph and assembles them
using an ILP solver.

Explanation:
Serves as the main prior baseline that the All‑At‑Once
approach seeks to improve upon and compare against.

strength: strong; depth: 1

221535103.c0

GPT‑f automated prover and proof assistant for
Metamath

Generative Language Modeling for Automated Theorem
Proving

The paper introduces GPT‑f, a system that couples
a decoder‑only Transformer (up to 774 M
parameters) with a custom Python Metamath
verifier. GPT‑f generates a proof‑step (theorem
identifier and substitution list) conditioned on a
Metamath goal, expands open goals by sampling
multiple tactics, and orders them by cumulative
log‑probability in a breadth‑first proof search.
The system also supports an interactive
proof‑assistant mode and an iterative self‑play
loop that trains a provability value function on
generated proof data, achieving state‑of‑the‑art
performance on the Metamath set.mm benchmark.

Prerequisite Edge Description

Description:
Transformer models that generate text autoregressively
using stacked self‑attention layers, as introduced in
the original Transformer paper and later scaled in
GPT‑2 and GPT‑3.

Explanation:
GPT‑f’s core generative component is a decoder‑only
Transformer; without this architecture the model could
not generate proof steps conditioned on goals.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Transformer models that generate text autoregressively
using stacked self‑attention layers, as introduced in
the original Transformer paper and later scaled in
GPT‑2 and GPT‑3.

Explanation:
GPT‑f’s core generative component is a decoder‑only
Transformer; without this architecture the model could
not generate proof steps conditioned on goals.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
An iterative procedure where the model generates
self‑play proof data, a provability value function is
trained on binary outcomes of intermediate goals, and
the improved value function guides subsequent proof
search.

Explanation:
GPT‑f’s self‑improvement loop that trains a
provability predictor V(g) and uses it to guide search
is based on this methodology.

strength: strong; depth: 1

218971783.c0

GPT-3 175‑billion‑parameter dense autoregressive transformer language model

Language Models are Few-Shot Learners

The paper introduces GPT‑3, a 175 B‑parameter
decoder‑only transformer trained on ~300 B tokens
of a filtered Common Crawl mixture. The model
scales depth, width, and compute roughly ten‑fold
beyond prior dense language models (e.g., GPT‑2)
while preserving the same autoregressive
architecture. GPT‑3 is evaluated without any
weight updates on a wide range of NLP tasks using
zero‑, one‑, and few‑shot prompting, achieving
performance competitive with state‑of‑the‑art
fine‑tuned systems on many benchmarks.

Prerequisite Edge Description

Description:
Observations that cross‑entropy loss and downstream
performance follow smooth power‑law trends with model
size, dataset size, and compute, enabling prediction
of returns from scaling.

Explanation:
Guided the decision to train a 175 B‑parameter model
and to expect improved few‑shot performance.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Observations that cross‑entropy loss and downstream
performance follow smooth power‑law trends with model
size, dataset size, and compute, enabling prediction
of returns from scaling.

Explanation:
Guided the decision to train a 175 B‑parameter model
and to expect improved few‑shot performance.

strength: strong; depth: 2

210861095.c0

Power-law scaling of Transformer language-model loss with model size, dataset size, and compute

Scaling Laws for Neural Language Models

The authors empirically demonstrate that the
cross-entropy loss of decoder-only Transformer
language models follows simple power-law
relationships with three independent scale
factors: the number of non-embedding parameters
(N), the number of training tokens (D), and the
total non-embedding compute used for training (C).
By training models ranging from 10^6 to >10^9
parameters on datasets spanning six orders of
magnitude, they fit equations (1.1)–(1.2) and
(1.5)–(1.6) that predict loss for any combination
of N, D, and C when the other two are not
bottlenecks. These scaling laws enable accurate
extrapolation of performance and inform optimal
allocation of compute resources.

210861095.c1

Closed-form loss model L(N,D) and universal overfitting scaling

Scaling Laws for Neural Language Models

The authors propose a compact analytical
expression L(N,D) (Equation 1.5) that predicts the
test cross‑entropy loss of decoder‑only
Transformer language models as a function of the
non‑embedding parameter count N and the dataset
size D when both are varied jointly. By fitting
this expression to a dense grid of experiments
spanning six orders of magnitude in N and D, they
demonstrate that the overfitting penalty follows a
simple power‑law dependence on the ratio
N^{0.74}/D, meaning that an eight‑fold increase in
model size requires only about a five‑fold
increase in data to avoid loss degradation. The
model accurately captures the full experimental
surface, providing a universal description of
overfitting across scales.

218971783.c7

Release of a family of smaller GPT-3 models

Language Models are Few-Shot Learners

The authors train eight additional GPT-3 models
ranging from 125 M to 13 B parameters using the
same dense decoder‑only transformer architecture,
data, and training pipeline as the 175 B model,
and publicly release the checkpoints. These models
enable the research community to perform detailed
scaling analyses and to apply language models in
settings where the full 175 B model is
computationally infeasible.

19449905.c5

Value‑network augmentation for MCTS using apprentice estimates

Thinking Fast and Slow with Deep Learning and Tree
Search

The paper introduces a multitask convolutional
neural network that jointly predicts a policy
distribution and a state‑value estimate for Hex
positions. The value target is obtained from
Monte‑Carlo rollouts of the current apprentice
policy, avoiding costly expert rollouts. During
tree search the learned value network replaces the
rollout component of Monte‑Carlo Tree Search,
providing fast value estimates that improve search
efficiency and increase playing strength.

Prerequisite Edge Description

Description:
The apprentice and value networks are trained by
minimising a Kullback‑Leibler divergence loss (for
policy) and a KL loss for value, using the Adam
stochastic optimisation algorithm.

Explanation:
These optimisation choices are necessary to
successfully learn the multitask network.

strength: strong; depth: 2

221535103.c1

Proof‑step conditional language‑modeling objective (GOAL → PROOFSTEP)

Generative Language Modeling for Automated Theorem
Proving

The paper defines a training objective that treats
each Metamath proof step as a conditional
language‑modeling problem: given a serialized goal
token sequence, a decoder‑only Transformer
autoregressively generates the full proof‑step
(theorem identifier and substitution list) that
resolves the goal. The objective is implemented as
a token‑level cross‑entropy loss on the
concatenated string "GOAL PROOFSTEP ". This
enables end‑to‑end learning of proof generation
without separate premise‑selection or substitution
modules and integrates directly with a standard
Transformer architecture.

211132980.c0

MetaGen: Neural forward‑generation system for Metamath theorems and proofs

Learning to Prove Theorems by Learning to Generate
Theorems

MetaGen is a neural generator that synthesizes new
Metamath theorems together with full proof trees
by performing forward reasoning over existing
theorems. It selects an invocable background
theorem, constructs a substitution using a
relevance network and a substitution network, and
optionally grafts existing proof subtrees to
obtain multi‑step proofs. The system enforces
constraints that prevent meaningless theorems
(e.g., hypotheses must be existing expressions)
and restricts invoked theorems to those that
appear as background facts in the training tasks,
thereby producing synthetic data that are useful
for training theorem provers.

Prerequisite Edge Description

Description:
Gated Recurrent Units (GRUs) are used to encode token
sequences representing Metamath expressions and to
decode substitution expressions. Bidirectional GRUs
provide contextual embeddings for the relevance and
substitution networks.

Explanation:
Both MetaGen and the prover rely on GRU
encoders/decoders to process symbolic inputs and
generate substitutions.

strength: strong; depth: 1

8242113.c1

Systematic comparison of neural architectures for embedding mathematical statements

DeepMath - Deep Sequence Models for Premise
Selection

The authors evaluate three families of neural
networks—pure convolutional networks, pure
recurrent LSTM/GRU networks, and hybrid
recurrent‑convolutional models—for encoding
variable‑length first‑order logic strings
extracted from the Mizar Mathematical Library.
Each architecture is trained on premise‑selection
data and its ability to rank relevant axioms for
new conjectures is measured. Results show that
shallow CNNs with max‑pooling consistently
outperform deeper recurrent models on this task,
providing the first large‑scale empirical analysis
of sequence‑model designs for formal mathematical
language.

Prerequisite Edge Description

Description:
An adaptive learning‑rate method that computes
individual learning rates for each parameter based on
estimates of first and second moments of the
gradients.

Explanation:
All neural models are trained using Adam, affecting
convergence and final performance.

strength: strong; depth: 2

3922816.c0

ARC Dataset: large‑scale grade‑school science QA benchmark with Challenge/Easy split

Think you have Solved Question Answering? Try ARC,
the AI2 Reasoning Challenge

The paper introduces the AI2 Reasoning Challenge
(ARC) dataset, a collection of 7,787
natural‑language multiple‑choice science questions
drawn from real standardized tests for grades 3‑9.
The questions are partitioned into an Easy Set
(5,197 items) and a Challenge Set (2,590 items)
based on whether two simple baseline solvers—an
information‑retrieval (IR) system and a
pointwise‑mutual‑information (PMI) co‑occurrence
system—answer them correctly. This split creates a
benchmark that isolates questions requiring deeper
reasoning and knowledge integration, providing a
new problem formulation for evaluating advanced QA
systems.

Prerequisite Edge Description

Description:
Prior reading‑comprehension and
natural‑language‑inference datasets (e.g., SQuAD,
SNLI, TriviaQA) together with high‑performing neural
models (e.g., BiDAF, Decomposable Attention) that
established the state of the art before ARC.

Explanation:
These works provide the experimental context and
motivate the need for a more reasoning‑heavy benchmark
such as ARC.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
Prior reading‑comprehension and
natural‑language‑inference datasets (e.g., SQuAD,
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models (e.g., BiDAF, Decomposable Attention) that
established the state of the art before ARC.

Explanation:
These works provide the experimental context and
motivate the need for a more reasoning‑heavy benchmark
such as ARC.

strength: strong; depth: 1

11816014.c0

Release of the Stanford Question Answering Dataset (SQuAD) v1.0

SQuAD: 100,000+ Questions for Machine
Comprehension of Text

The paper introduces SQuAD, a large-scale
reading‑comprehension benchmark comprising 107,785
crowd‑generated question–answer pairs over 536
Wikipedia articles. Each answer is a contiguous
text span from the passage, enabling
span‑selection evaluation rather than
multiple‑choice. The dataset is split into
training, development, and test sets and is freely
released for unrestricted research use, providing
both a resource dataset and a standardized
benchmark for machine reading comprehension.

Prerequisite Edge Description

Description:
Existing RC datasets such as MCTest, CNN/Daily Mail,
and the Children’s Book Test highlighted the need for
a large, high‑quality, span‑based benchmark.

Explanation:
The shortcomings of earlier datasets motivated the
creation of SQuAD and informed design choices (e.g.,
size, answer format).

strength: strong; depth: 2

Prerequisite Edge Description

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Existing RC datasets such as MCTest, CNN/Daily Mail,
and the Children’s Book Test highlighted the need for
a large, high‑quality, span‑based benchmark.

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Existing RC datasets such as MCTest, CNN/Daily Mail,
and the Children’s Book Test highlighted the need for
a large, high‑quality, span‑based benchmark.

Explanation:
The shortcomings of earlier datasets motivated the
creation of SQuAD and informed design choices (e.g.,
size, answer format).

strength: strong; depth: 2

2100831.c1

MCTest MC160 and MC500 Reading‑Comprehension Datasets with Standard Splits and Rich Metadata

**MCTest: A Challenge Dataset for the Open-Domain
Machine Comprehension of Text**

The authors construct and publicly release two
non‑copyrighted corpora, MC160 (160 story sets)
and MC500 (500 story sets), each consisting of a
150‑300 word fictional story, four multiple‑choice
questions with four answer options, correct answer
labels, and extensive metadata
(age‑appropriateness, grammar quality, clarity,
and a flag indicating whether a question requires
information from multiple sentences). The corpora
are accompanied by predefined train/dev/test
splits, enabling immediate benchmarking of
machine‑reading systems. By providing both the raw
data and the evaluation protocol, the resource
serves as a benchmark for open‑domain machine
comprehension research.

6203757.c0

Large-scale supervised reading-comprehension dataset (CNN/Daily Mail)

Teaching Machines to Read and Comprehend

The authors construct a corpus of roughly one
million (context, query, answer) triples by
harvesting news articles from CNN and the Daily
Mail together with their abstractive bullet‑point
summaries. Each bullet point is turned into a
Cloze‑style question by replacing a single entity
with a placeholder, after which all entities are
anonymised and randomly permuted. This yields a
realistic, large‑scale supervised resource for
training and evaluating machine‑reading models,
and formalises reading comprehension as the
conditional probability p(a | c,q).

6203757.c2

CNN/Daily Mail reading‑comprehension benchmark

Teaching Machines to Read and Comprehend

The authors define a standardized benchmark for
machine reading comprehension built on the CNN and
Daily Mail corpora. They publish fixed
train/validation/test splits, an evaluation
protocol that measures answer accuracy (p(a|c,q)),
and a suite of baseline results (majority,
frame‑semantic, word‑distance, Deep LSTM,
Attentive Reader, Impatient Reader). The benchmark
enforces document‑only reasoning by anonymising
and randomly permuting entity markers, thereby
eliminating world‑knowledge shortcuts.

14915449.c0

Children's Book Test (CBT) benchmark dataset

The Goldilocks Principle: Reading Children's Books
with Explicit Memory Representations

The paper introduces the Children’s Book Test, a
large‑scale benchmark built from freely available
children’s books on Project Gutenberg. Each
example consists of a 20‑sentence context and a
21st query sentence with one word removed; the
model must select the correct word from ten
candidate answers drawn from the context. The
benchmark is split into four word‑type
categories—named entities, common nouns, verbs,
and prepositions—to explicitly assess a model’s
ability to exploit wider discourse context for
semantically rich words, and provides public
training, validation, and test splits.

11816014.c4

Logistic regression model with lexicalized and dependency‑tree path features for span selection
on SQuAD

SQuAD: 100,000+ Questions for Machine
Comprehension of Text

The authors implement a feature‑based logistic
regression classifier to select answer spans in
the SQuAD reading‑comprehension task. The model
combines TF‑IDF word and bigram match scores,
root‑match indicators, span length and position
features, constituent label and POS‑tag features,
lexicalized word‑pair features, and
dependency‑tree path features extracted from
Stanford CoreNLP parses. Trained with AdaGrad and
L2 regularization, it attains 51.0 % F1 on the
SQuAD test set, substantially outperforming a
sliding‑window baseline, and an ablation study
shows that lexicalized and dependency‑path
features contribute the most to performance.

26501419.c0

TriviaQA large‑scale reading‑comprehension dataset with distant supervision

TriviaQA: A Large Scale Distantly Supervised
Challenge Dataset for Reading Comprehension

The paper releases TriviaQA, a publicly available
reading‑comprehension resource comprising over 650
K (question, answer, evidence‑document) triples.
Questions are authored by trivia enthusiasts
independent of any evidence, and evidence
documents are collected retrospectively from
Wikipedia and Web search results (average ≈6
documents per question). The release includes a
noisy automatically gathered portion, a clean
human‑verified subset of 1 975 triples, code for
data collection and preprocessing, baseline
experiments, and standard training, development,
and test splits for benchmarking RC models.

Prerequisite Edge Description

Description:
Prior reading‑comprehension datasets (e.g., SQuAD) and
their evaluation protocols (Exact Match, token‑level
F1) that define benchmark splits and performance
measures.

Explanation:
Provide the template for the train/dev/test splits and
the evaluation metrics adopted for TriviaQA.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Prior reading‑comprehension datasets (e.g., SQuAD) and
their evaluation protocols (Exact Match, token‑level
F1) that define benchmark splits and performance
measures.

Explanation:
Provide the template for the train/dev/test splits and
the evaluation metrics adopted for TriviaQA.

strength: strong; depth: 2

11816014.c5-1

Exact Match and token‑level F1 evaluation metrics
for span‑based QA

SQuAD: 100,000+ Questions for Machine
Comprehension of Text

The paper defines two token‑level evaluation
metrics for the SQuAD reading‑comprehension task.
Exact Match (EM) requires the predicted answer
span to match any ground‑truth span exactly, while
the token‑level F1 score measures the harmonic
mean of token precision and recall, ignoring
punctuation and articles. These metrics provide a
reproducible, fine‑grained benchmark for comparing
machine reading models on span selection.

26501419.c3-3

Modified BiDAF architecture for long,
multi-document evidence

TriviaQA: A Large Scale Distantly Supervised
Challenge Dataset for Reading Comprehension

The paper adapts the Bidirectional Attention Flow
(BiDAF) neural reading‑comprehension model to the
TriviaQA setting by (i) truncating each evidence
document to the first 800 tokens so that the model
can process long web and Wikipedia articles, (ii)
applying the model independently to each document
in the evidence set, and (iii) aggregating the
per‑document answer confidence scores via
summation to produce a final answer prediction.
This enables a state‑of‑the‑art RC architecture to
handle the noisy, multi‑document inputs
characteristic of TriviaQA.

Prerequisite Edge Description

Description:
A neural reading‑comprehension architecture that
encodes characters, words, and phrases, applies a
bidirectional attention flow between question and
context, and predicts answer start and end positions.

Explanation:
The modified system directly builds upon the BiDAF
architecture, requiring its original design and
implementation as a foundation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A neural reading‑comprehension architecture that
encodes characters, words, and phrases, applies a
bidirectional attention flow between question and
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The modified system directly builds upon the BiDAF
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Prerequisite Edge Description

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A neural reading‑comprehension architecture that
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context, and predicts answer start and end positions.

Explanation:
The modified system directly builds upon the BiDAF
architecture, requiring its original design and
implementation as a foundation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Token‑level metrics that compare predicted answer
spans to ground‑truth answers, measuring exact string
match and token overlap (F1).

Explanation:
The performance of the modified BiDAF model is
reported using these standard RC evaluation metrics.

strength: strong; depth: 2

8535316.c0

Hierarchical multi‑stage embedding pipeline

Bidirectional Attention Flow for Machine
Comprehension

The paper introduces a three‑level embedding
hierarchy applied to both context and query
tokens. First, each word is encoded with a
character‑level embedding obtained via a 1‑D
convolutional neural network (CNN) followed by
max‑pooling, capturing sub‑word morphology.
Second, a pretrained GloVe word vector is
concatenated to the character embedding, providing
lexical semantics. Third, the concatenated vectors
are processed by a two‑layer Highway network and a
bidirectional LSTM to produce contextual
embeddings that encode surrounding word
information before any attention is applied. This
pipeline yields rich token‑level features that
combine morphology, semantics, and context.

Prerequisite Edge Description

Description:
Dense word vectors learned on large corpora using the
Global Vectors (GloVe) method, providing semantic
representations for each token.

Explanation:
These embeddings constitute the second level of the
hierarchy, supplying lexical semantics that are
concatenated with character features.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Dense word vectors learned on large corpora using the
Global Vectors (GloVe) method, providing semantic
representations for each token.

Explanation:
These embeddings constitute the second level of the
hierarchy, supplying lexical semantics that are
concatenated with character features.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A feed‑forward network with gated shortcut connections
that transforms the concatenated character and word
embeddings before they are fed to the contextual LSTM,
facilitating deeper feature learning.

Explanation:
The Highway network is essential for effectively
combining the character‑level and word‑level vectors
prior to contextual encoding.

strength: strong; depth: 2

8535316.c1

Bidirectional attention flow layer with
memory‑less static attention

Bidirectional Attention Flow for Machine
Comprehension

The BiDAF model introduces an attention‑flow layer
that computes a similarity matrix S between every
context token and every query token using a
trainable scalar function α that combines
concatenated vectors, element‑wise products and
linear terms. From S it derives two complementary
attentions: (i) context‑to‑query (C2Q) attention,
producing a query‑aware vector for each context
word via a softmax over the query dimension, and
(ii) query‑to‑context (Q2C) attention, extracting
a single context vector by taking the max over the
context dimension followed by a softmax. The
attentions are memory‑less: each attention weight
depends only on the current context word and the
full query, allowing the attention vectors to be
pre‑computed and passed unchanged to a subsequent
modeling Bi‑LSTM, avoiding early summarization of
the context.

Prerequisite Edge Description

Description:
Soft attention aligns two sequences by computing a
similarity score for each pair of elements and
weighting one sequence with a softmax over the scores;
popularized in neural machine translation.

Explanation:
BiDAF’s similarity matrix S and the subsequent
softmax‑based weighting directly build on this
formulation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Dynamic attention models compute attention
recurrently, allowing the attention at step t to
depend on previous attention states (e.g., Match‑LSTM,
Memory Networks).

Explanation:
BiDAF’s contribution is defined by contrasting its
static, memory‑less attention with these dynamic
approaches.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Dynamic attention models compute attention
recurrently, allowing the attention at step t to
depend on previous attention states (e.g., Match‑LSTM,
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Prerequisite Edge Description

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Dynamic attention models compute attention
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depend on previous attention states (e.g., Match‑LSTM,
Memory Networks).

Explanation:
BiDAF’s contribution is defined by contrasting its
static, memory‑less attention with these dynamic
approaches.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
The attention‑over‑attention mechanism computes a
single context vector by taking the max over the
similarity matrix and applying a softmax, highlighting
the most relevant context words for the query.

Explanation:
BiDAF’s Q2C attention follows this formulation to
obtain the global context vector h.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Co‑attention models for VQA compute attention in both
directions (question→image and image→question) to
improve reasoning.

Explanation:
BiDAF’s use of both C2Q and Q2C attentions is
conceptually inspired by this bidirectional attention
paradigm.

strength: strong; depth: 2

8535316.c2

Task‑specific output head for QA formats

Bidirectional Attention Flow for Machine
Comprehension

BiDAF defines a modular output layer that is
decoupled from the shared encoder. For the SQuAD
span‑selection task the head consists of two
independent linear classifiers that produce
softmax distributions over start and end token
positions, followed by a joint maximisation to
select the answer span. For cloze‑style
reading‑comprehension tasks the head predicts only
a start index and applies a mask that restricts
scoring to anonymised entity tokens. This design
enables the same BiDAF encoder to be reused across
different machine‑comprehension formulations with
minimal changes.

Prerequisite Edge Description

Description:
Predicting answer spans as start and end token
positions using two independent linear classifiers
followed by a softmax over token positions and a joint
maximisation over all possible (start, end) pairs.

Explanation:
The SQuAD‑specific output head implements this
formulation, requiring the prior definition of
span‑selection as the standard answer format for the
SQuAD benchmark.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Predicting a single token (the missing anonymised
entity) by scoring only positions that correspond to
entity tokens and masking out all other tokens in the
softmax distribution.

Explanation:
The cloze‑specific output head relies on this
formulation to handle the CNN/DailyMail cloze task,
where answers are always anonymised entities.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Predicting a single token (the missing anonymised
entity) by scoring only positions that correspond to
entity tokens and masking out all other tokens in the
softmax distribution.

Explanation:
The cloze‑specific output head relies on this
formulation to handle the CNN/DailyMail cloze task,
where answers are always anonymised entities.

strength: strong; depth: 2

26501419.c5

Human‑verified clean evaluation subset of TriviaQA

TriviaQA: A Large Scale Distantly Supervised
Challenge Dataset for Reading Comprehension

The authors construct a high‑quality subset of
1,975 question‑document‑answer triples drawn from
the larger TriviaQA collection. Each triple is
manually verified: annotators confirm that the
evidence document contains all facts required to
answer the question, eliminating noise and
missing‑evidence issues present in the
automatically gathered data. This verified subset
is released alongside the full dataset and code,
providing a reliable benchmark for measuring true
reading‑comprehension performance.

8495258.c0

Decomposable Attention Model for Natural Language
Inference

A Decomposable Attention Model for Natural
Language Inference

The paper introduces a neural architecture for the
natural language inference (NLI) task that
replaces recurrent or convolutional encoders with
a three‑stage pipeline: (1) Attend – a soft,
dot‑product attention that aligns each word in the
premise with a weighted combination of words in
the hypothesis; (2) Compare – a feed‑forward
network that jointly processes each aligned word
pair; (3) Aggregate – sum‑pooling of the
comparison vectors followed by a final
feed‑forward classifier. The model operates
directly on fixed word‑embedding vectors, does not
rely on explicit word‑order information, and is
fully parallelizable across token positions,
achieving state‑of‑the‑art accuracy on SNLI with
far fewer parameters than LSTM‑based baselines.

Prerequisite Edge Description

Description:
Dense vector representations of words learned from
large corpora, providing a fixed‑dimensional semantic
space that can be used as input to neural models.

Explanation:
The Decomposable Attention Model operates directly on
word‑embedding vectors; without such representations
the model would have no input features.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A differentiable alignment technique that computes a
weighted sum of source representations for each target
token, typically using a compatibility function
followed by softmax normalization.

Explanation:
The Attend stage relies on soft‑attention to produce
the β and α aligned phrase vectors that drive the rest
of the pipeline.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A large, manually annotated dataset of 570K
premise‑hypothesis sentence pairs labeled for
entailment, contradiction, or neutral.

Explanation:
The model is designed and evaluated on SNLI; the
dataset provides the supervised signal needed to train
and benchmark the architecture.

strength: strong; depth: 2

8495258.c2

State‑of‑the‑Art Empirical Evaluation on SNLI with Parameter‑Efficient Decomposable Attention Models

A Decomposable Attention Model for Natural
Language Inference

The authors train the vanilla Decomposable
Attention (DA) model and its
intra‑sentence‑attention variant on the Stanford
Natural Language Inference (SNLI) corpus,
achieving 86.3% test accuracy with 382 K
parameters and 86.8% with 582 K parameters
respectively. These results surpass previously
reported LSTM, CNN, and tree‑based NLI systems
while using roughly an order of magnitude fewer
trainable parameters. The paper also provides a
per‑class accuracy breakdown and a qualitative
error analysis illustrating the strengths and
limitations of the approach.

Prerequisite Edge Description

Description:
A large, manually‑annotated dataset of 570 K
premise–hypothesis sentence pairs labeled for
entailment, contradiction, or neutral relations.

Explanation:
The experiments are conducted on this benchmark;
access to the data is essential for training and
evaluation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
300‑dimensional word vectors trained on global
word‑co‑occurrence statistics, later projected to 200
dimensions for the model.

Explanation:
The DA model uses fixed word embeddings as its input
representation, so these embeddings are a core
prerequisite.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Existing neural architectures for natural language
inference, including LSTM with attention (Rocktaschel
et al., 2016), LSTM (Wang & Jiang, 2016), and LSTMN
(Cheng et al., 2016).

Explanation:
The paper compares its results against these
baselines; understanding their performance is
necessary to contextualize the claimed improvements.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Existing neural architectures for natural language
inference, including LSTM with attention (Rocktaschel
et al., 2016), LSTM (Wang & Jiang, 2016), and LSTMN
(Cheng et al., 2016).

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et al., 2016), LSTM (Wang & Jiang, 2016), and LSTMN
(Cheng et al., 2016).

Explanation:
The paper compares its results against these
baselines; understanding their performance is
necessary to contextualize the claimed improvements.

strength: strong; depth: 2

2135897.c0

Conditional Encoding LSTM Architecture for RTE

Reasoning about Entailment with Neural Attention

The paper introduces a recurrent architecture for
Recognizing Textual Entailment (RTE) in which the
premise sentence is first processed by an LSTM;
the final cell state of this LSTM is then used to
initialise a second LSTM that reads the
hypothesis. By conditioning the hypothesis
encoding on the premise representation, the model
allows information to flow directly between the
two sentences, avoiding independent encoding of
each sentence. This design yields a
3.3‑percentage‑point accuracy improvement over the
independent‑encoding baseline on the Stanford
Natural Language Inference (SNLI) corpus.

6506243.c0

Long Short-Term Memory-Network (LSTMN)
architecture

Long Short-Term Memory-Networks for Machine
Reading

The paper proposes the Long Short-Term
Memory-Network (LSTMN), a recurrent neural network
that replaces the single memory cell of a standard
LSTM with an internal memory network composed of
multiple slots, each storing the contextual
representation of a token. The model maintains a
hidden-state tape for interaction with the
environment and a memory tape that grows with the
sequence length up to a predefined span, enabling
non-Markovian state updates. An intra-attention
mechanism softly addresses the memory slots at
each time step, allowing adaptive memory usage,
incremental left-to-right processing, and weak
induction of undirected lexical relations, thereby
mitigating the memory-compression and structural
limitations of vanilla LSTMs.

6506243.c3-3

Empirical evaluation of LSTMN with attention-fusion on SNLI natural language
inference

Long Short-Term Memory-Networks for Machine
Reading

The authors integrate the Long Short-Term
Memory-Network (LSTMN) into an encoder-decoder
framework using shallow and deep attention-fusion
strategies and evaluate the resulting models on
the Stanford Natural Language Inference (SNLI)
corpus. The shallow-fusion LSTMN and the
deep-fusion LSTMN (300‑dimensional hidden units)
achieve test accuracies of 84.3% and 86.3%
respectively, matching or surpassing prior
state‑of‑the‑art results while using a comparable
number of parameters. This demonstrates that
combining intra‑attention, inter‑sentence
attention, and a memory‑network‑augmented
recurrence effectively models sentence‑pair
semantics for natural language inference.

14786967.c0

Highway Network Architecture with Transform and Carry Gates

Highway Networks

The paper introduces Highway Networks, a
feed‑forward architecture in which each layer
computes a transformed representation H(x) and a
transform gate T(x). The layer output is a convex
combination y = H(x)·T(x) + x·(1‑T(x)), where the
carry gate C(x) is defined as 1‑T(x). This gating
mechanism creates "information highways" that
allow signals to propagate unchanged across many
layers, mitigating the vanishing‑gradient problem
and enabling the training of very deep networks
with standard stochastic gradient descent.

2926851.c0

Memory Networks conceptual framework

Memory Networks

The paper introduces a general class of learning
models called Memory Networks that combine a
long‑term external memory with standard inference
components. The architecture is defined by four
modular functions: I (input mapping), G (memory
update), O (output generation by retrieving
supporting memories), and R (response generation).
This high‑level framework is algorithm‑agnostic,
allowing any learning method to be used for each
module and enabling read/write access to a
scalable memory for tasks such as question
answering.

2926851.c1

Memory Neural Network (MemNN) architecture for
textual QA

Memory Networks

The paper presents a concrete neural
implementation of the Memory Networks framework
for question answering over text. Sentences (facts
or questions) are stored verbatim in a growing
memory array via the I and G modules. At inference
time the O module retrieves up to k supporting
memories using learned embedding similarity
scores, and the R module produces a single‑word
answer by scoring candidate answer words with a
separate embedding matrix. The implementation uses
bag‑of‑words feature mappings, distinct embedding
matrices for retrieval and answer scoring, and is
trained with a margin ranking loss optimized by
stochastic gradient descent.

5592690.c0

Boundary model: match-LSTM with Pointer Network for span prediction

Machine Comprehension Using Match-LSTM and Answer
Pointer

The paper introduces a neural architecture for
machine comprehension that predicts answer spans
by jointly applying a match-LSTM to attend a
passage to a question and a Pointer Network to
select the start and end token positions. The
match-LSTM encodes each passage token with
bidirectional contextual information and aligns it
to the question via attention, while the Answer
Pointer layer uses the pointer mechanism to output
two indices defining a contiguous answer span.
Evaluated on the SQuAD v1.1 benchmark, the
boundary model achieves an exact‑match score of
67.6% and an F1 of 77.0%, surpassing the
logistic‑regression baseline.

5592690.c1

Sequence model: match-LSTM with Pointer Network for token‑sequence generation

Machine Comprehension Using Match-LSTM and Answer
Pointer

The paper introduces a neural architecture that
combines a match‑LSTM encoder with a Pointer
Network decoder to generate answers as a
variable‑length sequence of token positions from
the passage. The decoder emits indices into the
passage and includes a special end‑of‑answer
token, allowing the model to select
non‑consecutive tokens. The model is trained
end‑to‑end on the SQuAD v1.1 dataset and evaluated
against baselines, showing lower performance than
the span‑based boundary model but demonstrating
the feasibility of sequence‑based answer
generation.

9205021.c0

Attention-over-Attention (AoA) Reader Architecture

Attention-over-Attention Neural Networks for
Reading Comprehension

The paper introduces the Attention-over-Attention
(AoA) Reader, a neural network for cloze‑style
reading comprehension. The model encodes document
and query tokens with a shared bidirectional GRU,
computes a pairwise matching matrix via
dot‑product, derives a query‑to‑document attention
for each query word (column‑wise softmax) and a
document‑to‑query attention (row‑wise softmax),
averages the latter to obtain a global query‑level
attention vector, and finally produces an
"attended attention" over the document by taking
the dot product of the two attention matrices. The
resulting document‑level attention scores are
summed over all occurrences of each candidate word
to yield answer probabilities, removing the need
for heuristic merging functions while adding
negligible computational overhead.

868693.c1

Co‑Attention Mechanism for Joint Image‑Question Reasoning

Hierarchical Question-Image Co-Attention for
Visual Question Answering

The paper introduces a co‑attention framework that
simultaneously computes attention distributions
over image regions and over question tokens,
allowing the two modalities to guide each other.
An affinity matrix is built between image feature
vectors and hierarchical question representations,
and attention weights are learned from this
matrix. Two concrete strategies are defined: (a)
parallel co‑attention, which generates image and
question attentions in a single step, and (b)
alternating co‑attention, which iteratively
refines one modality’s attention conditioned on
the other’s summary. This symmetric design enables
richer cross‑modal interaction than prior
visual‑only attention models.

6203757.c1

Entity anonymisation and permutation pipeline

Teaching Machines to Read and Comprehend

The paper introduces a preprocessing pipeline for
the reading‑comprehension dataset that (a) runs a
coreference resolver to group mentions of the same
real‑world entity, (b) replaces each entity with
an abstract marker such as ent23, and (c) randomly
permutes these markers each time a data point is
loaded. This removes any reliance on
world‑knowledge cues and forces models to answer
questions solely from the provided document
context.

8535316.c3-1

State‑of‑the‑art performance on SQuAD benchmark

Bidirectional Attention Flow for Machine
Comprehension

The BiDAF model attains the highest reported Exact
Match (73.3 %) and F1 (81.1 %) scores on the
hidden test set of the Stanford Question Answering
Dataset (SQuAD) at the time of submission, using
an ensemble of 12 independently trained runs. This
result demonstrates that the proposed
architecture, which combines hierarchical
embeddings, memory‑less bidirectional attention
flow, and a span‑selection output head,
outperforms all previously published systems on
this large‑scale reading‑comprehension benchmark.

Prerequisite Edge Description

Description:
A three‑level embedding stack that combines
character‑level CNN embeddings, pre‑trained GloVe word
vectors, and contextual bidirectional LSTM
representations passed through a Highway network.

Explanation:
Rich token representations are required for the
downstream attention and modeling layers to achieve
high accuracy on SQuAD.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A three‑level embedding stack that combines
character‑level CNN embeddings, pre‑trained GloVe word
vectors, and contextual bidirectional LSTM
representations passed through a Highway network.

Explanation:
Rich token representations are required for the
downstream attention and modeling layers to achieve
high accuracy on SQuAD.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
The Stanford Question Answering Dataset (over 100 k
questions) with evaluation based on Exact Match and
token‑level F1 scores.

Explanation:
The dataset defines the task and the metrics against
which the model’s performance is measured.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
The Stanford Question Answering Dataset (over 100 k
questions) with evaluation based on Exact Match and
token‑level F1 scores.

Explanation:
The dataset defines the task and the metrics against
which the model’s performance is measured.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Training on a single Titan X GPU using the AdaDelta
optimizer, dropout regularization, and moving‑average
weight tracking.

Explanation:
Effective training procedures are required to achieve
the reported performance levels.

strength: strong; depth: 2

8535316.c3-2

State‑of‑the‑art performance on CNN/DailyMail
cloze test

Bidirectional Attention Flow for Machine
Comprehension

Using a slight modification of the output layer,
BiDAF achieves the highest reported accuracy on
the CNN and DailyMail cloze‑style
reading‑comprehension datasets for both single‑run
and 12‑model ensemble configurations, surpassing
prior memory‑network and attention‑based
approaches. This demonstrates the model’s ability
to generalize from span‑selection to
entity‑prediction tasks. The experiments reuse the
same BiDAF encoder as for SQuAD, are trained with
AdaDelta and dropout on a single GPU, and evaluate
on the standard CNN/DailyMail benchmark.

Prerequisite Edge Description

Description:
A three‑level token representation that combines
character‑level CNN embeddings, pretrained GloVe word
vectors, and contextual BiLSTM outputs after a Highway
network. This pipeline supplies rich lexical and
contextual features to downstream attention and
modeling components.

Explanation:
Accurate token representations are essential for the
attention flow and modeling layers to correctly
identify answer entities in the cloze task.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A three‑level token representation that combines
character‑level CNN embeddings, pretrained GloVe word
vectors, and contextual BiLSTM outputs after a Highway
network. This pipeline supplies rich lexical and
contextual features to downstream attention and
modeling components.

Explanation:
Accurate token representations are essential for the
attention flow and modeling layers to correctly
identify answer entities in the cloze task.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Large‑scale news article datasets where a named‑entity
placeholder is removed from a summary sentence; the
task is to predict the missing entity from the article
text.

Explanation:
These datasets define the evaluation benchmark on
which the reported state‑of‑the‑art results are
measured.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Large‑scale news article datasets where a named‑entity
placeholder is removed from a summary sentence; the
task is to predict the missing entity from the article
text.

Explanation:
These datasets define the evaluation benchmark on
which the reported state‑of‑the‑art results are
measured.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Training uses the AdaDelta optimizer with a learning
rate of 0.5, dropout (rate 0.2) on CNNs and LSTMs,
moving‑average weight tracking, and runs on a single
Titan X GPU for ~20 hours per model.

Explanation:
Effective optimization and regularization are
necessary to achieve the high accuracy reported on the
cloze benchmarks.

strength: strong; depth: 2

218974301.c0

WorldTree V2 multi‑fact explanation dataset

WorldTree V2: A Corpus of Science-Domain
Structured Explanations and Inference Patterns
supporting Multi-Hop Inference

The authors release WorldTree V2, a corpus of
5,114 standardized elementary and middle‑school
science multiple‑choice questions each paired with
a detailed, human‑authored explanation graph that
combines on average six atomic facts. Explanations
are represented as lexically‑connected graphs
drawn from a manually curated semi‑structured
knowledge base of 9,216 facts across 66 tables.
This resource triples the size of the original
WorldTree corpus and covers roughly half of all
publicly available U.S. science exam questions,
providing training and evaluation data for
many‑fact multi‑hop inference.

Prerequisite Edge Description

Description:
The AI2 Reasoning Challenge (ARC) dataset comprising
7,787 multiple‑choice science questions collected from
U.S. state exams, covering grades 3‑9.

Explanation:
WorldTree V2 questions are drawn from the ARC corpus,
providing the raw question set for which explanations
are authored; thus it is a peripheral prerequisite.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A hierarchical classification of science exam
questions into fine‑grained curriculum topics, later
aggregated into nine coarse categories, used to
organize and analyze the question set.

Explanation:
The authors sort ARC questions using this annotation
to ensure topical consistency during explanation
authoring; it is a peripheral prerequisite.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
A hierarchical classification of science exam
questions into fine‑grained curriculum topics, later
aggregated into nine coarse categories, used to
organize and analyze the question set.

Explanation:
The authors sort ARC questions using this annotation
to ensure topical consistency during explanation
authoring; it is a peripheral prerequisite.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
The original WorldTree resource containing 1,680
explanation graphs and a supporting tablestore, along
with the annotation protocol introduced by Jansen et
al. (2018).

Explanation:
WorldTree V2 builds directly on the design, annotation
guidelines, and knowledge base of the original
WorldTree, making it a core prerequisite.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
The original WorldTree resource containing 1,680
explanation graphs and a supporting tablestore, along
with the annotation protocol introduced by Jansen et
al. (2018).

Explanation:
WorldTree V2 builds directly on the design, annotation
guidelines, and knowledge base of the original
WorldTree, making it a core prerequisite.

strength: strong; depth: 1

Prerequisite Edge Description

Description:
The original WorldTree resource containing 1,680
explanation graphs and a supporting tablestore, along
with the annotation protocol introduced by Jansen et
al. (2018).

Explanation:
WorldTree V2 builds directly on the design, annotation
guidelines, and knowledge base of the original
WorldTree, making it a core prerequisite.

strength: strong; depth: 1

199668863.c0

Hierarchical taxonomy of 406 fine-grained problem
domains for K‑12 science exams

Multi-class Hierarchical Question Classification
for Multiple Choice Science Exams

The authors construct a six‑level hierarchical
taxonomy covering 406 detailed problem domains
(e.g., Astronomy → Gravitational Pull, Life
Science → Photosynthesis). The taxonomy is derived
from the NY Regents exam syllabus, three publicly
available science study guides, and the 7,787
multiple‑choice questions in the ARC corpus. Each
node is accompanied by a concise textual
definition, enabling categorization of questions
at multiple granularities for downstream
question‑classification and QA systems.

Prerequisite Edge Description

Description:
A collection of 7,787 multiple‑choice science
questions from U.S. 3rd‑ to 9th‑grade exams, used to
ground and validate the taxonomy.

Explanation:
The taxonomy was expanded and validated by analyzing
the content and distribution of the ARC questions,
ensuring coverage of real exam items.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A seminal taxonomy for open‑domain question
classification that defines coarse and fine‑grained
answer types and hierarchical feature propagation.

Explanation:
The authors used Li and Roth’s taxonomy as a
methodological template for building a multi‑level
hierarchy and for propagating predictions across
levels.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Procedures for double annotation, resolution of
disagreements, and measurement of agreement using
Cohen’s Kappa.

Explanation:
High‑quality taxonomy construction required reliable
human annotation; understanding and applying Kappa
statistics was necessary to assess and improve label
consistency.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
Procedures for double annotation, resolution of
disagreements, and measurement of agreement using
Cohen’s Kappa.

Explanation:
High‑quality taxonomy construction required reliable
human annotation; understanding and applying Kappa
statistics was necessary to assess and improve label
consistency.

strength: strong; depth: 2

11039301.c1

Two‑Layered Semantic Taxonomy of Answer Types

Learning Question Classifiers

The paper defines a two‑level taxonomy for answer
types used in open‑domain question answering. Six
coarse categories (ABBREVIATION, ENTITY,
DESCRIPTION, HUMAN, LOCATION, NUMERIC VALUE) are
introduced to align with existing entity
recognizers and to provide a manageable top‑level
partition. Each coarse class is further refined
into 50 mutually exclusive fine‑grained subclasses
(e.g., city, speed, definition) based on analysis
of the TREC‑10 question set, yielding a natural
semantic hierarchy that serves as the label space
for a hierarchical classifier and guides
downstream answer‑selection strategies.

5421278.c1

Comprehensive exposition of Cohen's kappa
statistic

Interrater reliability: the kappa statistic

The paper presents a detailed exposition of
Cohen's kappa, the chance‑adjusted interrater
reliability coefficient for two raters. It derives
the full mathematical formulation, defining
observed agreement Pr(a) and expected agreement
Pr(e) from marginal totals of a 2×2 contingency
table, and shows step‑by‑step computation. The
exposition also recounts Jacob Cohen’s original
1960s development of the statistic and restates
the conventional qualitative interpretation scale
(≤0 no agreement, 0.01‑0.20 slight, …, 0.81‑1.00
almost perfect).

5421278.c4

Guideline to report both percent agreement and Cohen's kappa based on expected guessing

Interrater reliability: the kappa statistic

The authors recommend that researchers calculate
and present both the simple percent agreement
statistic and Cohen's kappa when evaluating
interrater reliability. They advise using percent
agreement when raters are well‑trained and random
guessing is unlikely, and using kappa when
substantial guessing is suspected, thereby
providing a practical decision rule that balances
interpretability with adjustment for chance
agreement.

199668863.c1

Annotated ARC question‑classification dataset (7,787 questions)

Multi-class Hierarchical Question Classification
for Multiple Choice Science Exams

The authors release a publicly available dataset
that augments the 7,787 multiple‑choice science
questions from the ARC corpus with up to two
hierarchical labels drawn from a 406‑category
taxonomy. Each question was independently labeled
by two expert annotators, disagreements were
resolved, and the final labels achieve an
estimated 96 % accuracy (Cohen's κ = 0.58 at the
finest level). The release includes the full
taxonomy, label definitions, and serves both as a
resource and a benchmark for hierarchical,
multi‑label question classification.

Prerequisite Edge Description

Description:
The AI2 Reasoning Challenge (ARC) dataset supplies

7,787 multiple‑choice science questions from U.S. 3rd‑ to 9th‑grade exams, which serve as the raw material

for annotation.

Explanation:
Without the original ARC question set there would be
no items to label, making it a core prerequisite for
the dataset creation.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A statistical measure of agreement between annotators,
with interpretive guidelines for moderate to strong
agreement (McHugh, 2012).

Explanation:
Used to quantify and report the reliability of the
annotation process, informing the resolution step and
overall label quality.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A statistical measure of agreement between annotators,
with interpretive guidelines for moderate to strong
agreement (McHugh, 2012).

Explanation:
Used to quantify and report the reliability of the
annotation process, informing the resolution step and
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strength: strong; depth: 2

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overall label quality.

strength: strong; depth: 2

Prerequisite Edge Description

Description:
A statistical measure of agreement between annotators,
with interpretive guidelines for moderate to strong
agreement (McHugh, 2012).

Explanation:
Used to quantify and report the reliability of the
annotation process, informing the resolution step and
overall label quality.

strength: strong; depth: 2

5421278.c2-1

Revised minimum Cohen's kappa threshold for acceptable interrater reliability in health
research

Interrater reliability: the kappa statistic

The authors propose that health‑related studies
should require a Cohen's kappa of at least 0.80 to
be considered acceptable, arguing that the
traditional "moderate" band (0.40–0.60) can
conceal 40–50 % erroneous data. A new
interpretation table is presented that
reclassifies kappa values below 0.80 as
insufficient for clinical decision‑making.

5421278.c3

Guidelines for 95% confidence interval computation for Cohen's kappa and recommended minimum sample
sizes

Interrater reliability: the kappa statistic

The paper provides a practical procedure for
constructing a 95 % confidence interval around a
Cohen’s κ estimate using the normal‑approximation
formula κ ± 1.96·SE(κ). It explains how the
standard error of κ depends on the number of
paired ratings and demonstrates that small samples
yield wide intervals. Based on this relationship,
the authors recommend a minimum of 30 paired
observations to achieve a reasonably narrow
interval and suggest that samples of 1 000 or more
produce very precise estimates. These guidelines
link sample‑size planning directly to the
reliability assessment of κ in health‑research
contexts.

5421278.c6-1

Critical analysis of the rater independence assumption underlying Cohen's kappa

Interrater reliability: the kappa statistic

The paper argues that the assumption of
statistical independence between raters, which
underlies the derivation of Cohen's kappa, is
frequently violated in health‑care rating tasks
because raters may be trained together, share
protocols, or otherwise influence each other's
judgments, leading to biased kappa estimates.

5421278.c6-2

Critical analysis of the marginal‑total chance‑agreement assumption in Cohen's kappa

Interrater reliability: the kappa statistic

The paper critiques the use of marginal totals of
the rating matrix to estimate chance agreement,
noting that this estimate presumes raters behave
as independent random draws from the marginal
distribution—a condition that may be violated when
categories are unequally prevalent or when raters
share systematic biases.

5421278.c6-3

Critical analysis of the guessing‑behavior assumption in Cohen's kappa

Interrater reliability: the kappa statistic

The paper examines the assumption that when raters
guess, the probability of each possible rating
equals the marginal proportion of that rating in
the data, arguing that real‑world guessing may
follow different patterns, which can cause the
kappa adjustment for chance to be inaccurate.

3623373.c0

WorldTree explanation graph corpus

WorldTree: A Corpus of Explanation Graphs for
Elementary Science Questions supporting Multi-Hop
Inference

The paper introduces WorldTree, an open‑source
dataset comprising 1,680 elementary‑science
multiple‑choice questions (grades 3‑5) each paired
with a multi‑sentence explanation graph.
Explanations are provided both as free‑text
sentences and as a lexically‑overlapping graph of
table rows, with each sentence assigned one of
four roles (central, grounding, background,
lexical glue). The resource also includes the
underlying semi‑structured tablestore and the
annotation tooling used to construct the corpus,
enabling training and evaluation of explainable
question‑answering systems.

Prerequisite Edge Description

Description:
Research demonstrating that current QA systems lack
human‑readable justifications and that aggregating
multiple knowledge pieces is challenging, highlighting
the need for a gold‑standard explanation corpus.

Explanation:
These studies establish the need for a curated
explanation resource and guide the overall project
goal.

strength: strong; depth: 2

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Prerequisite Edge Description

Description:
Existing table‑store resources such as the Khashabi et
al. (2016) tablestore that demonstrated the utility of
semi‑structured knowledge for QA and informed the
design of the new tablestore.

Explanation:
These earlier resources showed feasibility of
table‑based reasoning and guided the construction of
the WorldTree tablestore.

strength: strong; depth: 2

2498076.c0

Explanation‑centered analysis framework for elementary science QA

What's in an Explanation? Characterizing Knowledge
and Inference Requirements for Elementary Science
Exams

The paper introduces a systematic framework that
constructs gold natural‑language explanations for
multiple‑choice elementary science questions by
retrieving grade‑appropriate sentences from study
guides, a science dictionary, and Simple English
Wiktionary, reusing sentences when possible and
inserting bridge sentences that explicitly link
concepts. These explanations are annotated with a
fine‑grained taxonomy of knowledge and inference
types and are used to characterize the knowledge
and inference requirements of each question,
enabling quantitative comparison of
retrieval‑based and inference‑based solvers and
informing future knowledge‑base construction and
solver design.

2498076.c5

Analytical insights on knowledge distribution and implications for QA system design

What's in an Explanation? Characterizing Knowledge
and Inference Requirements for Elementary Science
Exams

The authors analyze the fine‑grained knowledge and
inference requirements of 212 elementary science
questions, finding that roughly 77 % of them
require complex inference involving multiple
knowledge types. They then compare a
retrieval‑only learning‑to‑rank model with two
inference‑based solvers (TableILP and STITCH) on
the same dataset, showing that inference solvers
achieve higher overall accuracy and especially
improve performance on the complex‑inference
subset. From these observations they argue that
future QA systems should employ ensembles or
question‑specific solver selection and that the
fine‑grained taxonomy can guide the construction
of targeted knowledge bases.

7145148.c4

Semi‑structured table corpus for elementary‑science QA

Question Answering via Integer Programming over
Semi-Structured Knowledge

The authors construct a knowledge base comprising
65 manually or semi‑automatically built tables
(≈5,000 rows) whose schemas are derived from
elementary‑science curricula, together with four
Open‑IE tables (≈2,600 rows) extracted from
science texts. Each table is represented as an
n‑ary predicate over short natural‑language
phrases, providing a semi‑structured substrate for
the TableILP reasoning system. The corpus is
released publicly as a resource for QA research on
elementary‑science multiple‑choice questions.

3623373.c5

Guidelines for explanation depth and annotation protocol

WorldTree: A Corpus of Explanation Graphs for
Elementary Science Questions supporting Multi-Hop
Inference

The paper defines a set of design principles that
specify four levels of explanatory depth (young
child, domain novice, domain expert, first
principles) and couples these with a detailed
annotation protocol. The protocol includes
explicit sentence‑role labeling (central,
grounding, background, lexical glue), a multi‑pass
review process, and reuse strategies to ensure
that explanations are consistent, high‑quality,
and suitable for training explainable inference
models for elementary science QA.

Prerequisite Edge Description

Description:
Prior work that quantified the number of knowledge
pieces needed per question, identified common
inference types, and introduced the notion of
explanation depth for elementary science QA.

Explanation:
Provides the empirical basis for estimating how many
knowledge pieces must be aggregated and informs the
design of depth levels and annotation granularity.

strength: strong; depth: 2

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3623373.c3

Web‑based collaborative annotation tool for
constructing explanation graphs

WorldTree: A Corpus of Explanation Graphs for
Elementary Science Questions supporting Multi-Hop
Inference

The authors built a web‑based collaborative
interface that enables annotators to author
explanation graphs for elementary science
questions. The tool provides keyword search over
the semi‑structured tablestore, re‑use of
previously authored explanations, visualisation of
lexical overlap between sentences, and flagging of
quality issues. Explanations are stored as ordered
lists of table‑row identifiers, allowing later
modifications to the underlying tables to
propagate automatically to all explanations. The
interface also implements a multi‑pass review
workflow with role‑labeling flags to enforce the
annotation protocol.

233297051.c0

Formulation of explanation as multistep entailment
trees

Explaining Answers with Entailment Trees

The paper reconceives open-domain textual
question-answering explanation as the construction
of a directed entailment tree, where each node is
a multi-premise textual entailment step that
incrementally derives the hypothesis (question +
answer) from known facts. This formulation
separates the correctness of the derivation from
its utility for users, enabling systematic
evaluation of explanations independent of answer
selection. By defining explanations as trees
rather than isolated rationales, the work provides
a high-level conceptual framework that can guide
future research on explainable QA.

Prerequisite Edge Description

Description:
Multi-premise textual entailment extends standard
natural-language inference to infer a hypothesis from
two or more premises, requiring models to combine
information across sentences.

Explanation:
The entailment-tree formulation relies on each node
being a valid multi-premise entailment step.

strength: strong; depth: 0

Prerequisite Edge Description

Description:
Existing QA explanation methods provide short textual
rationales or supporting sentences rather than a full
chain of reasoning.

Explanation:
These works motivate the need for a richer, structured
explanation format and are contrasted with the
entailment-tree approach.

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Prerequisite Edge Description

Description:
Transformer-based language models can generate formal
or natural-language proofs by sequentially applying
inference rules, demonstrating the feasibility of
end-to-end generation of structured reasoning.

Explanation:
The EntailmentWriter model and the linearisation
scheme are directly inspired by proof-generation
systems such as ProofWriter.

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Prerequisite Edge Description

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Datasets such as ARC and WorldTree provide science
questions together with a corpus of domain-specific
facts, establishing a setting where multi-step
reasoning is required.

Explanation:
The formulation is applied to open-domain QA and uses
these datasets as the source of facts and as
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